

NKPC HSA Bayesian State-Space Estimation: Unemployment gap

June 23, 2026

1 Data and Transformations

Raw data are read from `data/raw/`. Processed model-ready data are written to `data/processed/`. The HSA competition aggregator convention is

$$N_t^{model} = \frac{100 \log(N_t^{level}) - \overline{100 \log(N^{level})}}{10}.$$

Thus N_t^{model} , $\hat{N}_t$, and $\bar{N}_t$ are centered and measured in ten-log-point units. If a run metadata file records another `n_transform`, interpret the corresponding coefficients using that metadata.

2 Model Specifications

The CES benchmark is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + \kappa x_t + e_t.$$

The dynamic HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + \kappa x_t - \theta \hat{N}_t + e_t.$$

The steady HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + (\kappa_0 + \delta \bar{N}_t) x_t + e_t.$$

The both-effects HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + (\kappa_0 + \delta \bar{N}_t) x_t - (\theta_0 + \gamma \bar{N}_t) \hat{N}_t + e_t.$$

The maintained HSA decomposition is

$$N_t = \hat{N}_t + \bar{N}_t, \quad \hat{N}_t = \rho_1 \hat{N}_{t-1} + \rho_2 \hat{N}_{t-2} + u_t, \quad \bar{N}_t = n + \bar{N}_{t-1} + \epsilon_t.$$

3 Unit and Covariance Conventions

Priors for κ , κ_0 , and δ are specified in physical units. Legacy Gibbs samplers internally use `KAPPA_SCALE=100` when the design matrix contains $x_t/100$. All posterior draws, tables, figures, SDDR calculations, and model-comparison summaries are reported in physical units. The default dynamic-HSA shock covariance allows only (e_t, ζ_t) correlation; diagonal and historical full covariance modes are available only through explicit run metadata. Because the default N transform is centered and divided by 10, the reported δ , θ , θ_0 , and γ coefficients are already effects for a ten-log-point movement in the corresponding N component. κ_0 and θ_0 are intercepts at average $\bar{N}_t$.

4 Coefficient Summary

Table 1 reports posterior means and 95% credible intervals for key parameters across all four models (baseline prior, full sample). Cells show *mean* [*ci_2.5*, *ci_97.5*].

Table 1: Posterior means and 95% credible intervals by model

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.647 [0.543, 0.749]	0.641 [0.543, 0.742]	0.594 [0.474, 0.708]	0.613 [0.484, 0.731]
kappa	0.056 [-0.008, 0.120]	NaN	0.085 [-0.011, 0.191]	NaN
kappa_0	NaN	0.091 [0.013, 0.176]	NaN	0.077 [-0.011, 0.174]
delta	NaN	0.029 [0.001, 0.056]	NaN	0.018 [-0.026, 0.052]
theta	NaN	NaN	0.237 [-0.266, 0.619]	NaN
theta_0	NaN	NaN	NaN	0.120 [-0.206, 0.494]
gamma	NaN	NaN	NaN	-0.006 [-0.044, 0.035]
rho_1	NaN	0.788 [0.416, 1.082]	0.917 [0.625, 1.141]	0.956 [0.709, 1.193]
rho_2	NaN	0.002 [-0.288, 0.270]	0.012 [-0.189, 0.228]	-0.037 [-0.267, 0.182]
phi_1	0.949 [0.893, 1.006]	0.949 [0.892, 1.004]	0.950 [0.898, 1.004]	0.950 [0.894, 1.004]
lambda_ez	0.042 [-0.167, 0.255]	0.026 [-0.187, 0.235]	NaN	0.022 [-0.188, 0.226]
n	NaN	-0.030 [-0.092, 0.031]	-0.015 [-0.068, 0.029]	-0.026 [-0.087, 0.035]

5 Result Blocks

Results are organized by condition. Each section fixes a data specification, prior specification, and sample period, then reports the coefficient table, prior/posterior overlay by model, Savage-Dickey density ratios, time-varying κ_t path, and model-comparison output for that condition.

6 unemployment_gap: prior=baseline, period=full

Condition. data=unemployment_gap, prior=baseline, period=full, constraint=unrestricted (unrestricted).

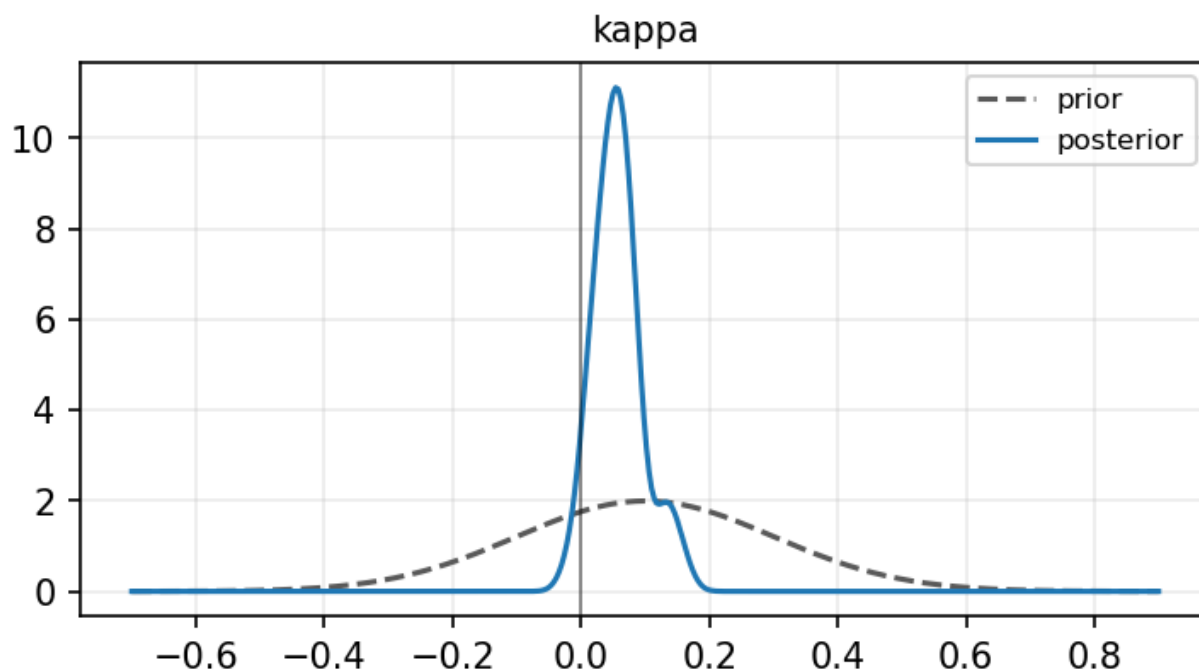
6.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.643 [0.549, 0.762]	0.658 [0.549, 0.758]	0.592 [0.521, 0.663]	0.634 [0.533, 0.744]
kappa	0.058 [0.006, 0.140]	NaN	0.094 [0.044, 0.160]	NaN
kappa_0	NaN	0.076 [-0.367, 0.366]	NaN	-0.059 [-0.370, 0.217]
delta	NaN	-0.000 [-0.002, 0.002]	NaN	0.001 [-0.001, 0.003]
theta	NaN	NaN	0.012 [0.006, 0.018]	NaN
theta_0	NaN	NaN	NaN	0.026 [-0.392, 0.434]
gamma	NaN	NaN	NaN	0.000 [-0.002, 0.002]
rho_1	NaN	1.813 [1.714, 1.901]	1.872 [1.790, 1.941]	1.194 [1.005, 1.392]
rho_2	NaN	-0.817 [-0.905, -0.716]	-0.876 [-0.945, -0.792]	-0.272 [-0.472, -0.068]
phi_1	0.953 [0.891, 0.996]	0.953 [0.907, 0.990]	0.955 [0.912, 1.000]	0.950 [0.915, 1.002]
lambda_ez	0.028 [-0.137, 0.182]	0.023 [-0.148, 0.188]	NaN	0.019 [-0.170, 0.220]
n	NaN	-0.061 [-0.175, 0.065]	0.001 [-0.091, 0.097]	-0.090 [-0.181, 0.037]

6.2 Prior vs Posterior by Model

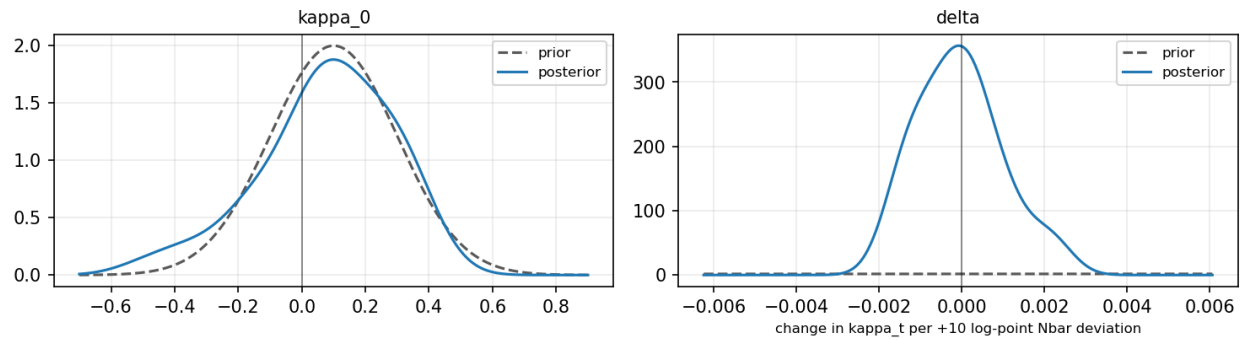
ces

Prior vs posterior — ces



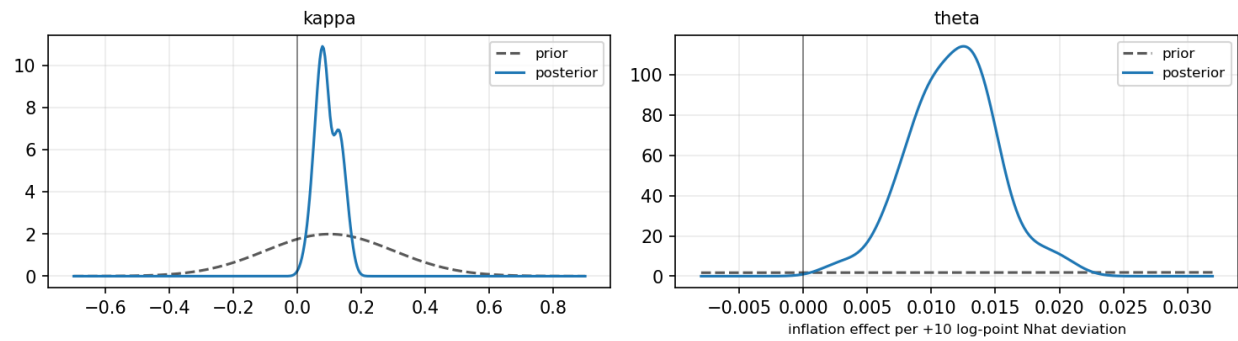
hsa_steady

Prior vs posterior — hsa_steady



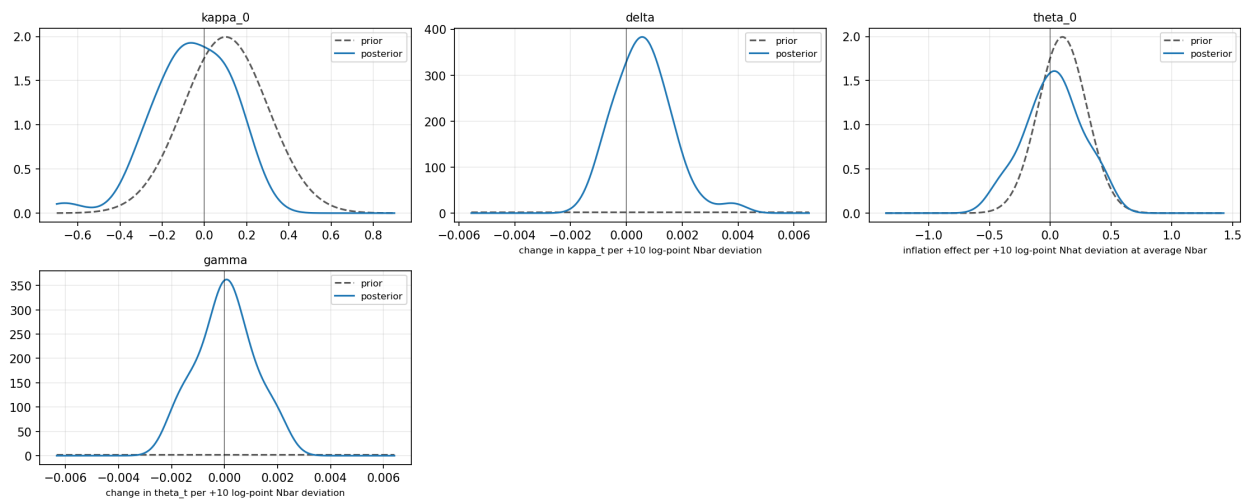
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

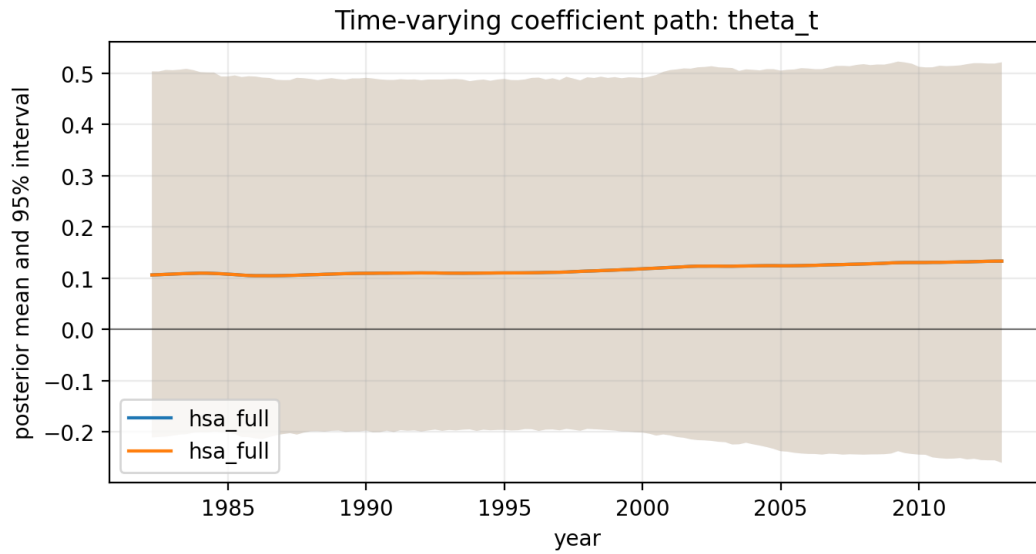
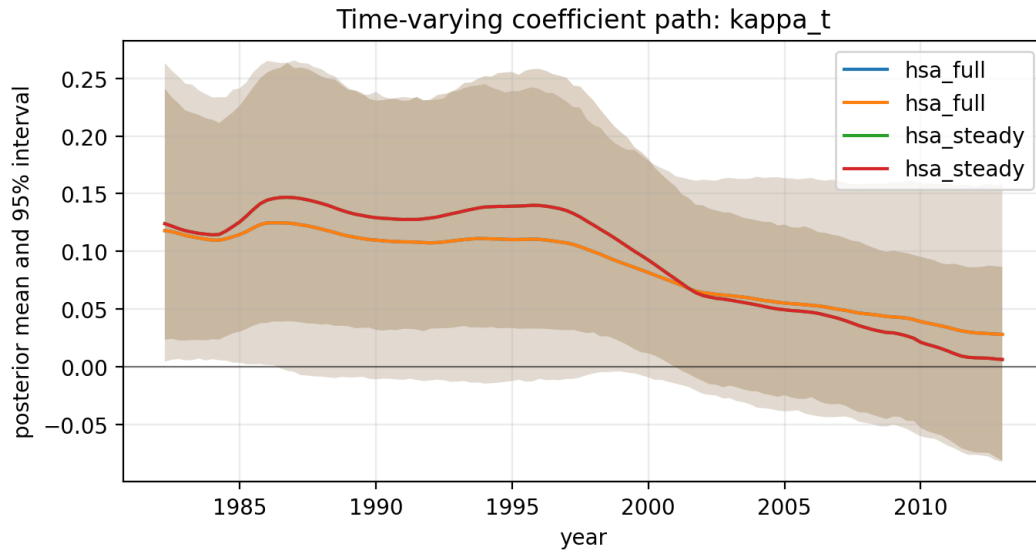
Prior vs posterior — hsa_full



6.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	1.6329
ces	kappa=0	0.1000	0.2000	1.6329
hsa_dynamic	kappa=0	0.1000	0.2000	1.1462
hsa_dynamic	theta=0	0.1000	0.2000	0.2501
hsa_dynamic	kappa=0	0.1000	0.2000	1.1462
hsa_dynamic	theta=0	0.1000	0.2000	0.2501
hsa_full	kappa_0=0	0.1000	0.2000	1.3799
hsa_full	delta=0	0.0000	0.0200	0.5268
hsa_full	theta_0=0	0.1000	0.2000	1.3570
hsa_full	gamma=0	0.0000	0.0200	0.8911
hsa_full	kappa_0=0	0.1000	0.2000	1.3799
hsa_full	delta=0	0.0000	0.0200	0.5268
hsa_full	theta_0=0	0.1000	0.2000	1.3570
hsa_full	gamma=0	0.0000	0.0200	0.8911
hsa_steady	kappa_0=0	0.1000	0.2000	0.4568
hsa_steady	delta=0	0.0000	0.0200	0.1589
hsa_steady	kappa_0=0	0.1000	0.2000	0.4568
hsa_steady	delta=0	0.0000	0.0200	0.1589

6.4 Time-Varying Kappa Path



6.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-119.9413	0.6354	-249.0143	1.0000	NaN	NaN	NaN
ces	NaN	NaN	-249.0143	1.0000	NaN	NaN	NaN
hsa_dynamic	-114.1660	0.6076	NaN	NaN	NaN	0.2501	NaN
hsa_dynamic	NaN	NaN	NaN	NaN	NaN	0.2501	NaN
hsa_full	-115.9013	0.6157	NaN	NaN	0.5268	NaN	0.8911
hsa_full	-115.9013	0.6157	NaN	NaN	0.5268	NaN	0.8911
hsa_steady	-115.6612	0.6134	-308.1333	0.0000	0.1589	NaN	NaN
hsa_steady	NaN	NaN	-308.1333	0.0000	0.1589	NaN	NaN

7 unemployment_gap: prior=baseline, period=full, constraint=kappa $i=0$

Condition. data=unemployment_gap, prior=baseline, period=full, constraint=restricted_kappa (kappa $i=0$).

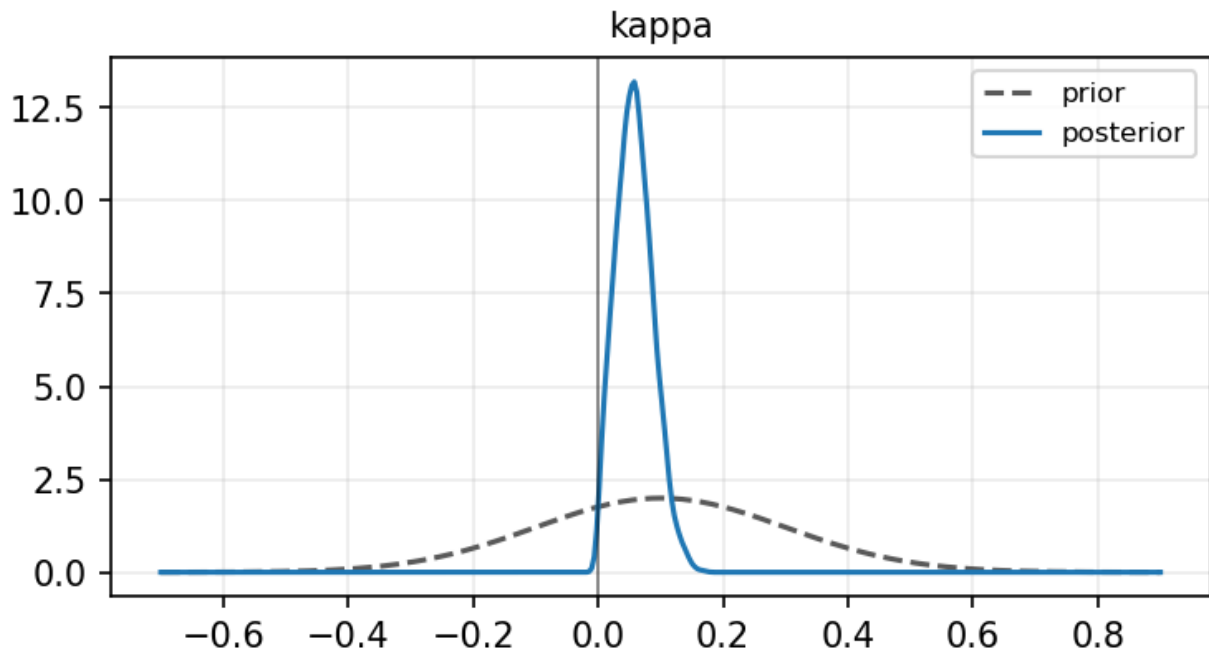
7.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.644 [0.540, 0.749]	0.641 [0.543, 0.742]	0.593 [0.473, 0.707]	0.613 [0.484, 0.731]
kappa	0.058 [0.007, 0.121]	NaN	0.087 [0.008, 0.189]	NaN
kappa_0	NaN	0.091 [0.013, 0.176]	NaN	0.077 [-0.011, 0.174]
delta	NaN	0.029 [0.001, 0.056]	NaN	0.018 [-0.026, 0.052]
theta	NaN	NaN	0.234 [-0.258, 0.632]	NaN
theta_0	NaN	NaN	NaN	0.120 [-0.206, 0.494]
gamma	NaN	NaN	NaN	-0.006 [-0.044, 0.035]
rho_1	NaN	0.788 [0.416, 1.082]	0.919 [0.652, 1.147]	0.956 [0.709, 1.193]
rho_2	NaN	0.002 [-0.288, 0.270]	0.014 [-0.198, 0.222]	-0.037 [-0.267, 0.182]
phi_1	0.950 [0.895, 1.004]	0.949 [0.892, 1.004]	0.951 [0.898, 1.004]	0.950 [0.894, 1.004]
lambda_ez	0.039 [-0.168, 0.243]	0.026 [-0.187, 0.235]	NaN	0.022 [-0.188, 0.226]
n	NaN	-0.030 [-0.092, 0.031]	-0.015 [-0.069, 0.030]	-0.026 [-0.087, 0.035]

7.2 Prior vs Posterior by Model

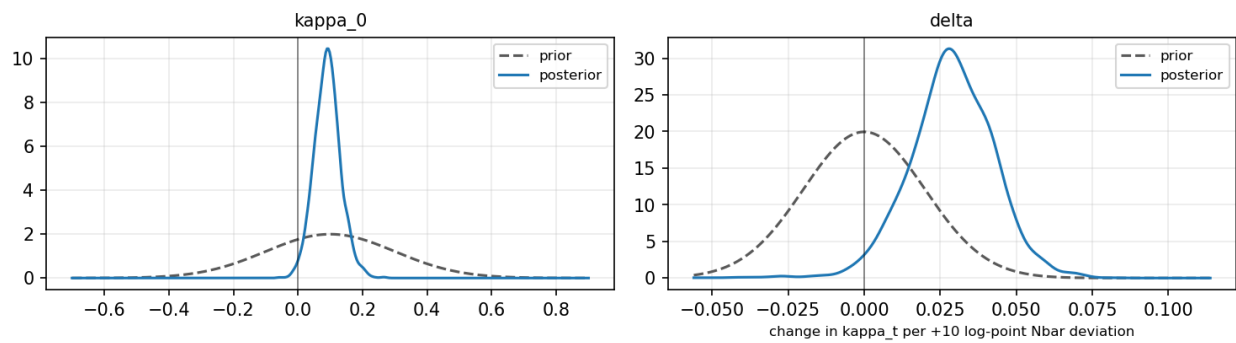
ces

Prior vs posterior — ces



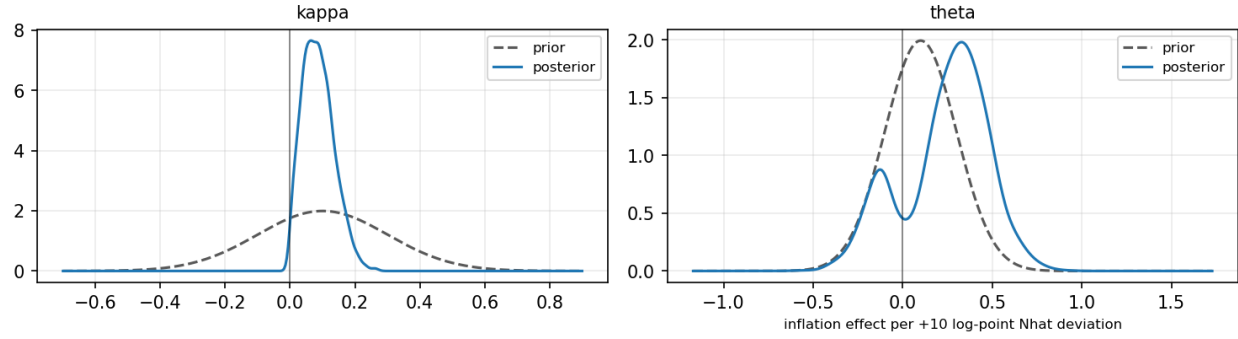
hsa_steady

Prior vs posterior — hsa_steady



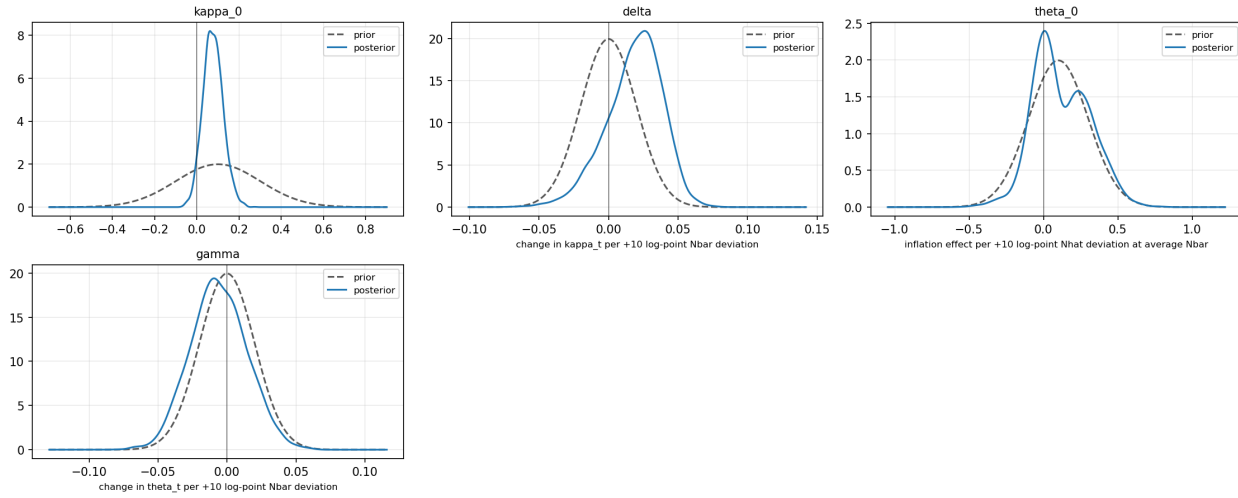
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

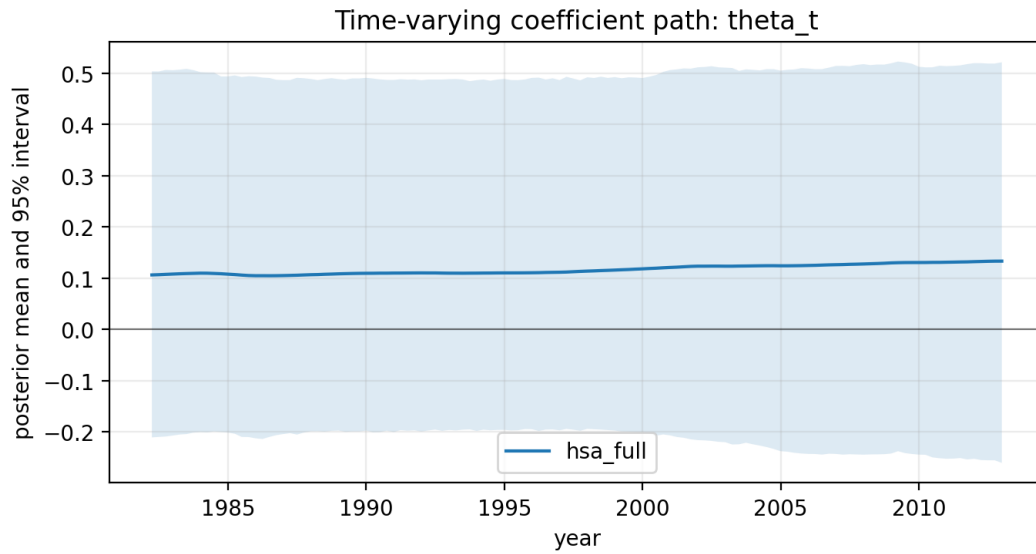
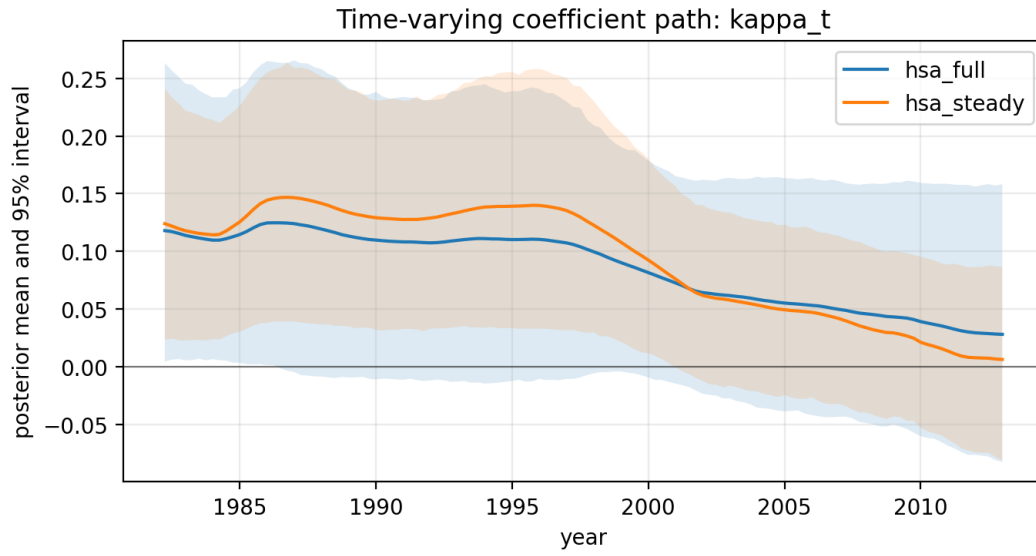
Prior vs posterior — hsa_full



7.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	0.9967
hsa_dynamic	kappa=0	0.1000	0.2000	0.9267
hsa_dynamic	theta=0	0.1000	0.2000	0.2589
hsa_full	kappa_0=0	0.1000	0.2000	1.3799
hsa_full	delta=0	0.0000	0.0200	0.5268
hsa_full	theta_0=0	0.1000	0.2000	1.3570
hsa_full	gamma=0	0.0000	0.0200	0.8911
hsa_steady	kappa_0=0	0.1000	0.2000	0.4568
hsa_steady	delta=0	0.0000	0.0200	0.1589

7.4 Time-Varying Kappa Path



7.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-119.9344	0.6354	-249.0224	1.0000	NaN	NaN	NaN
hsa_dynamic	-114.3149	0.6083	NaN	NaN	NaN	0.2589	NaN
hsa_full	-115.9013	0.6157	NaN	NaN	0.5268	NaN	0.8911
hsa_steady	-115.6612	0.6134	-308.1333	0.0000	0.1589	NaN	NaN

8 unemployment_gap: prior=baseline, period=pre_2008

Condition. data=unemployment_gap, prior=baseline, period=pre_2008, constraint=unrestricted (unrestricted).

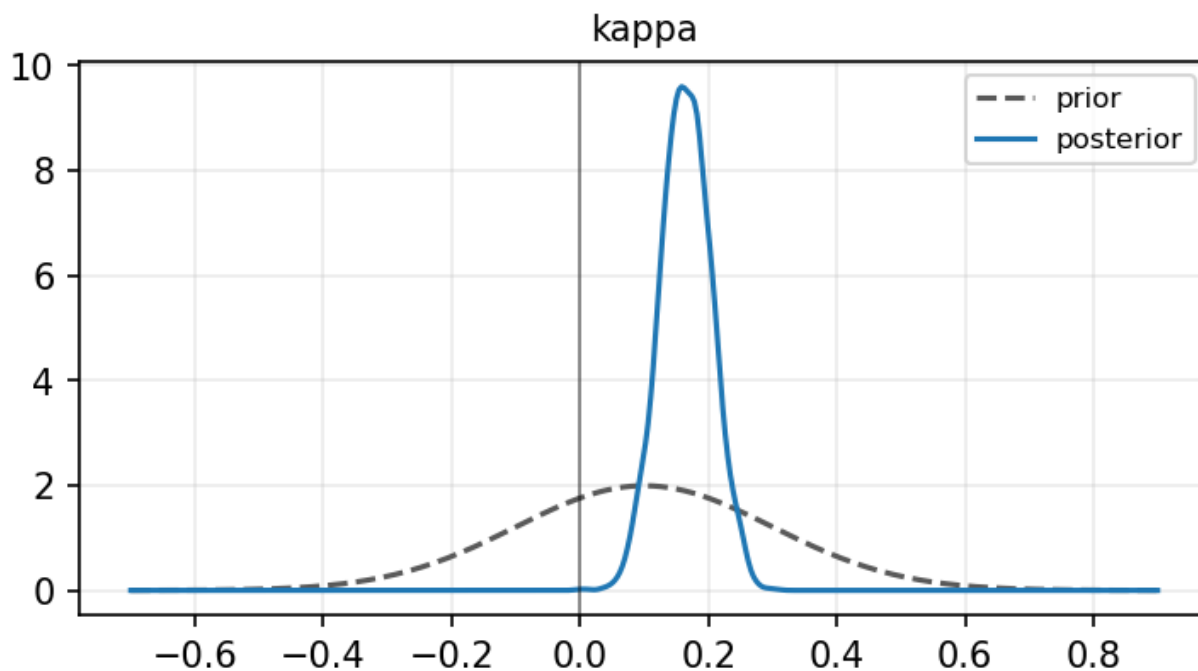
8.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.680 [0.572, 0.783]	0.670 [0.591, 0.765]	0.633 [0.503, 0.753]	0.614 [0.487, 0.742]
kappa	0.165 [0.087, 0.243]	NaN	0.180 [0.075, 0.290]	NaN
kappa_0	NaN	0.098 [-0.192, 0.462]	NaN	0.178 [0.052, 0.311]
delta	NaN	0.000 [-0.002, 0.002]	NaN	0.001 [-0.035, 0.037]
theta	NaN	NaN	0.292 [-0.169, 0.680]	NaN
theta_0	NaN	NaN	NaN	0.259 [0.019, 0.526]
gamma	NaN	NaN	NaN	-0.000 [-0.040, 0.038]
rho_1	NaN	1.812 [1.681, 1.905]	0.820 [0.424, 1.102]	0.950 [0.688, 1.202]
rho_2	NaN	-0.818 [-0.911, -0.686]	-0.027 [-0.284, 0.197]	-0.063 [-0.298, 0.168]
phi_1	0.894 [0.815, 0.972]	0.890 [0.823, 0.960]	0.896 [0.821, 0.970]	0.895 [0.814, 0.974]
lambda_ez	-0.062 [-0.265, 0.138]	-0.083 [-0.194, 0.069]	NaN	-0.089 [-0.285, 0.104]
n	NaN	-0.027 [-0.150, 0.130]	-0.015 [-0.064, 0.032]	-0.014 [-0.076, 0.050]

8.2 Prior vs Posterior by Model

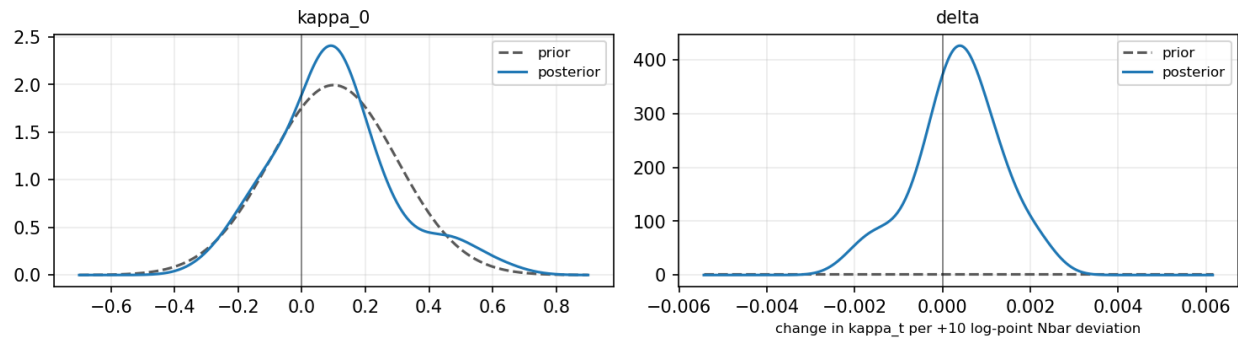
ces

Prior vs posterior — ces



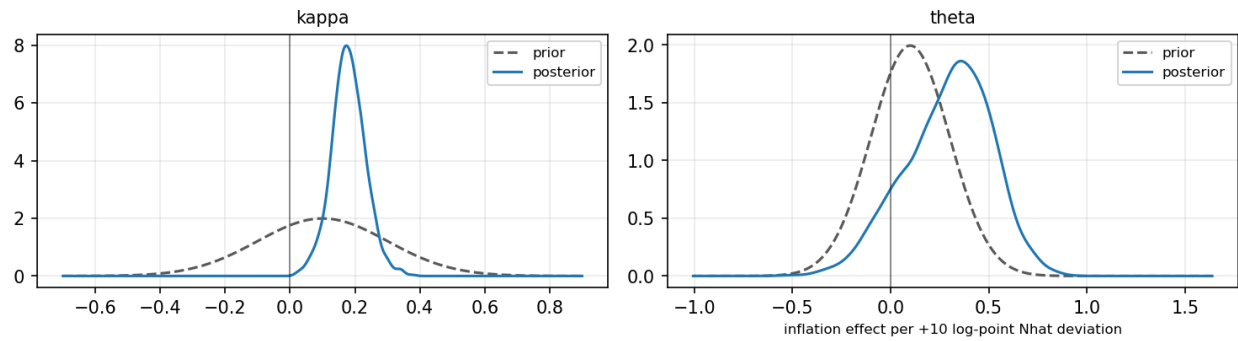
hsa_steady

Prior vs posterior — hsa_steady



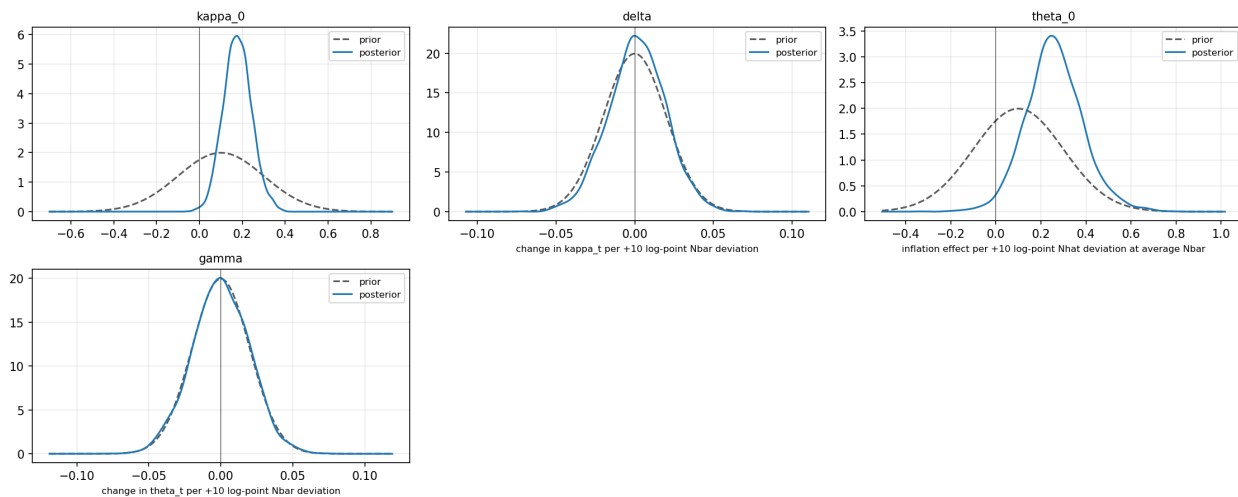
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

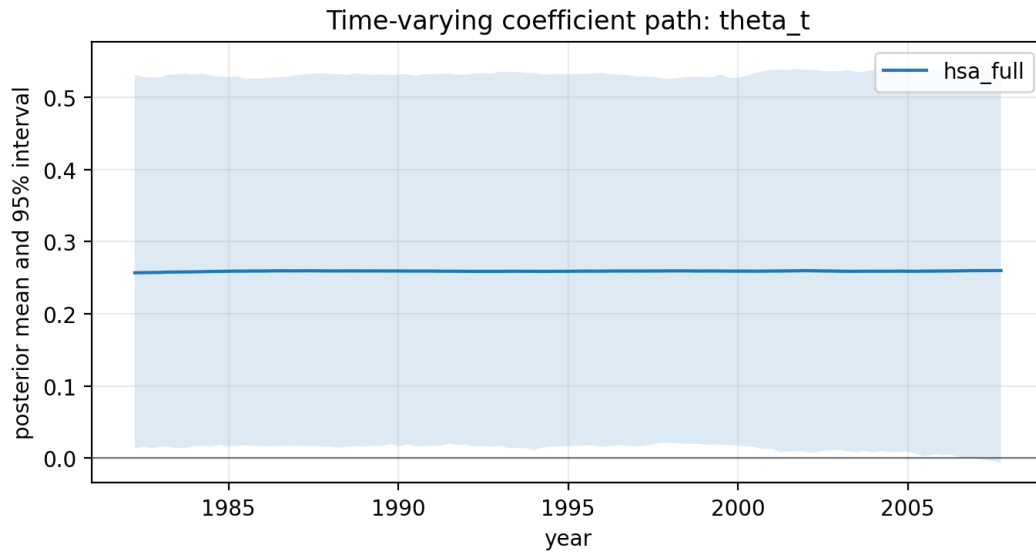
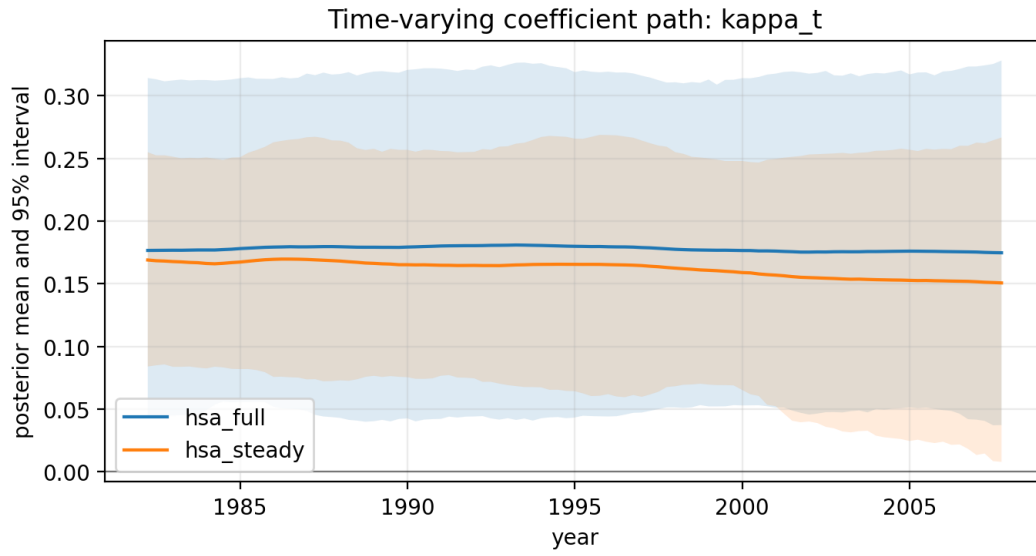
Prior vs posterior — hsa_full



8.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	0.0125
hsa_dynamic	kappa=0	0.1000	0.2000	0.0071
hsa_dynamic	theta=0	0.1000	0.2000	0.4281
hsa_full	kappa_0=0	0.1000	0.2000	0.0864
hsa_full	delta=0	0.0000	0.0200	1.1131
hsa_full	theta_0=0	0.1000	0.2000	0.1872
hsa_full	gamma=0	0.0000	0.0200	1.0049
hsa_steady	kappa_0=0	0.1000	0.2000	0.0235
hsa_steady	delta=0	0.0000	0.0200	0.9575

8.4 Time-Varying Kappa Path



8.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-69.1468	0.4700	-174.3969	1.0000	NaN	NaN	NaN
hsa_dynamic	-62.3895	0.4432	NaN	NaN	NaN	0.4281	NaN
hsa_full	-60.1745	0.4272	NaN	NaN	1.1131	NaN	1.0049
hsa_steady	-68.8480	0.4686	-236.6890	0.0000	0.9575	NaN	NaN

9 unemployment_gap: prior=baseline, period=post_2008

Condition. data=unemployment_gap, prior=baseline, period=post_2008, constraint=unrestricted (unrestricted).

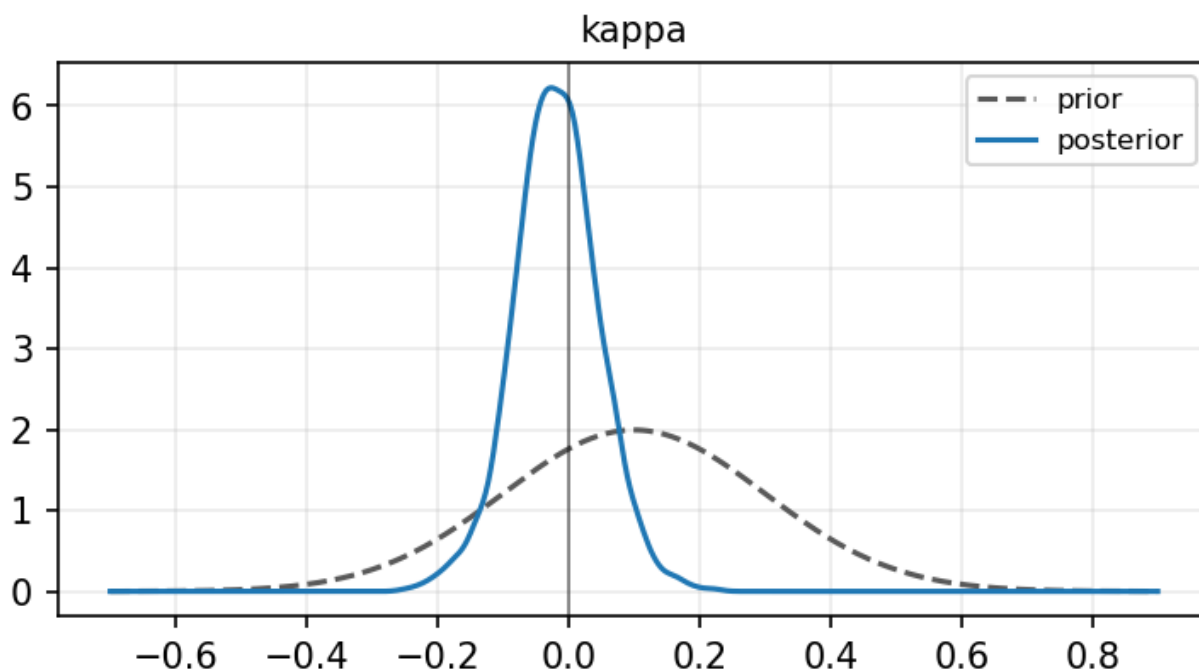
9.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.573 [0.326, 0.807]	0.610 [0.409, 0.789]	0.588 [0.365, 0.812]	0.574 [0.324, 0.803]
kappa	-0.019 [-0.149, 0.108]	NaN	-0.022 [-0.133, 0.086]	NaN
kappa_0	NaN	0.067 [-0.425, 0.464]	NaN	-0.022 [-0.147, 0.109]
delta	NaN	-0.000 [-0.003, 0.002]	NaN	0.001 [-0.039, 0.041]
theta	NaN	NaN	0.158 [-0.360, 0.623]	NaN
theta_0	NaN	NaN	NaN	0.104 [-0.307, 0.509]
gamma	NaN	NaN	NaN	-0.000 [-0.040, 0.038]
rho_1	NaN	0.533 [0.112, 0.912]	0.532 [0.134, 0.914]	0.580 [0.176, 0.970]
rho_2	NaN	-0.445 [-0.727, -0.196]	-0.351 [-0.705, -0.020]	-0.355 [-0.720, 0.015]
phi_1	0.976 [0.874, 1.074]	0.984 [0.862, 1.070]	0.983 [0.915, 1.046]	0.975 [0.873, 1.070]
lambda_ez	0.286 [-0.273, 0.844]	0.248 [-0.263, 0.623]	NaN	0.254 [-0.334, 0.819]
n	NaN	-0.064 [-0.215, 0.186]	-0.005 [-0.162, 0.156]	-0.001 [-0.174, 0.175]

9.2 Prior vs Posterior by Model

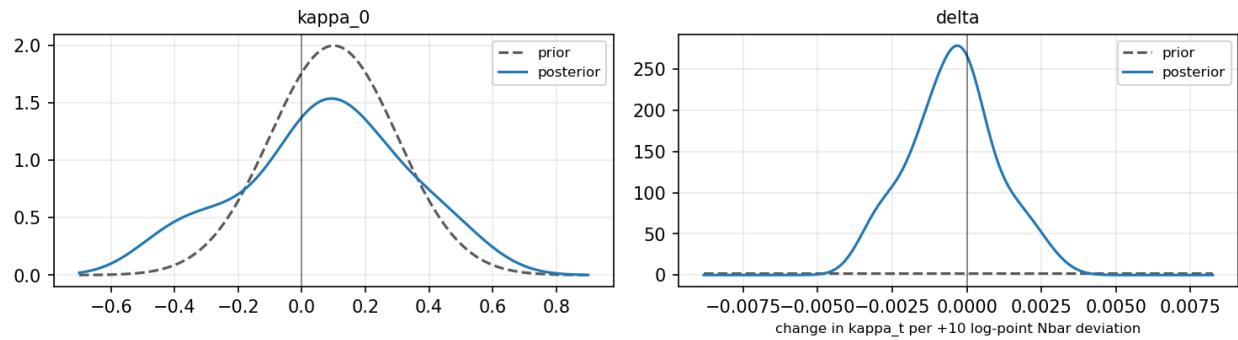
ces

Prior vs posterior — ces



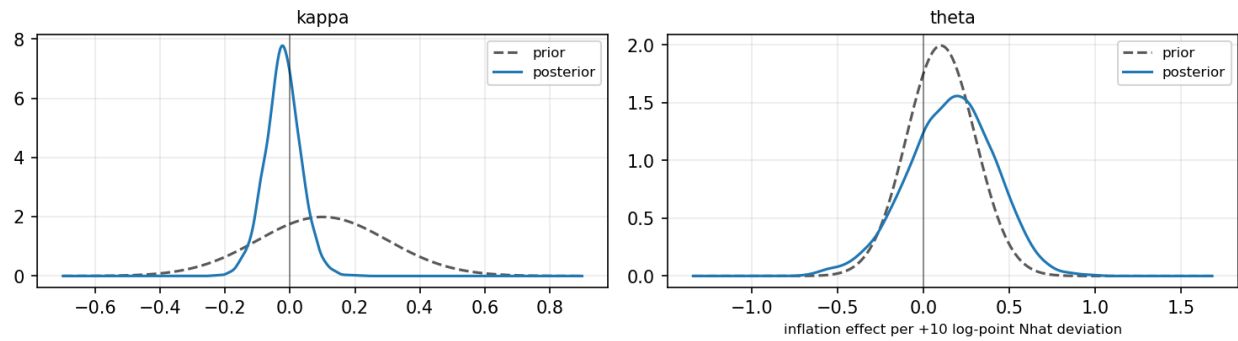
hsa_steady

Prior vs posterior — hsa_steady



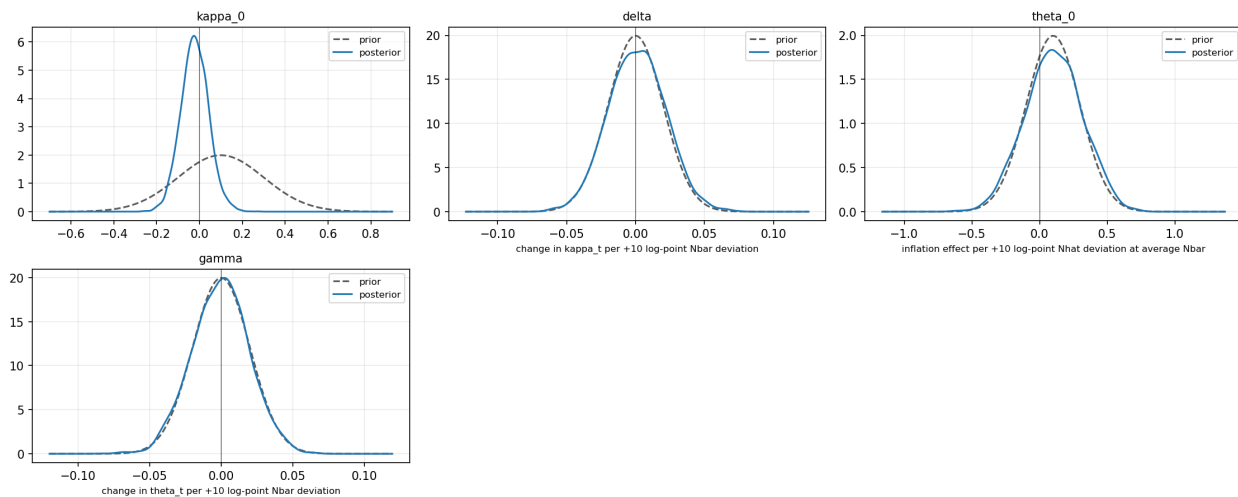
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

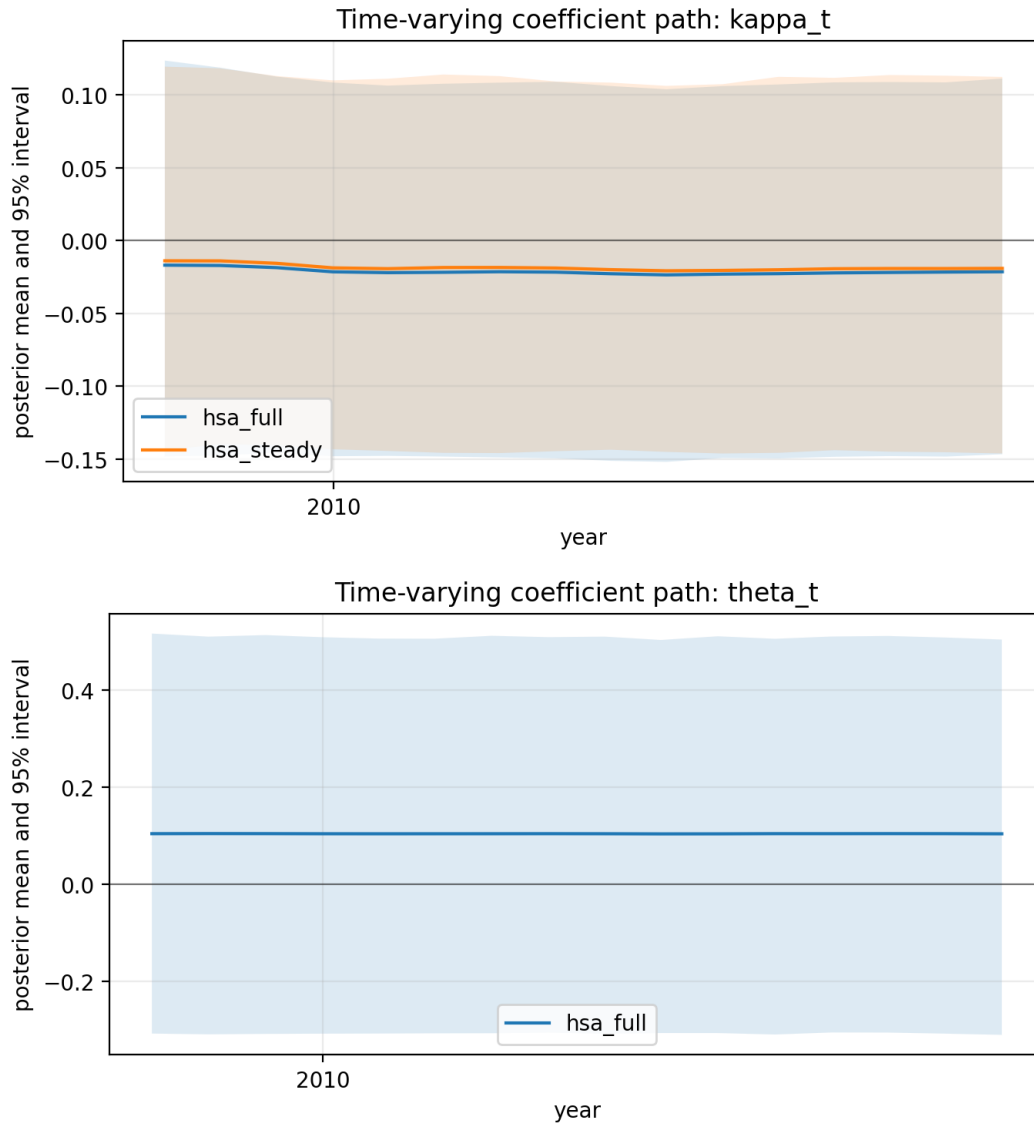
Prior vs posterior — hsa_full



9.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	3.4449
hsa_dynamic	kappa=0	0.1000	0.2000	3.8731
hsa_dynamic	theta=0	0.1000	0.2000	0.7088
hsa_full	kappa_0=0	0.1000	0.2000	3.2427
hsa_full	delta=0	0.0000	0.0200	0.9060
hsa_full	theta_0=0	0.1000	0.2000	0.9383
hsa_full	gamma=0	0.0000	0.0200	0.9925
hsa_steady	kappa_0=0	0.1000	0.2000	3.3503
hsa_steady	delta=0	0.0000	0.0200	0.9636

9.4 Time-Varying Kappa Path



9.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-21.7346	0.9270	-51.7157	1.0000	NaN	NaN	NaN
hsa_dynamic	-21.2531	0.9076	NaN	NaN	NaN	0.7088	NaN
hsa_full	-21.4511	0.9129	NaN	NaN	0.9060	NaN	0.9925
hsa_steady	-21.6396	0.9197	-75.7537	0.0000	0.9636	NaN	NaN

10 unemployment_gap: prior=baseline, period=exclude_covid

Condition. data=unemployment_gap, prior=baseline, period=exclude_covid, constraint=unrestricted (unrestricted).

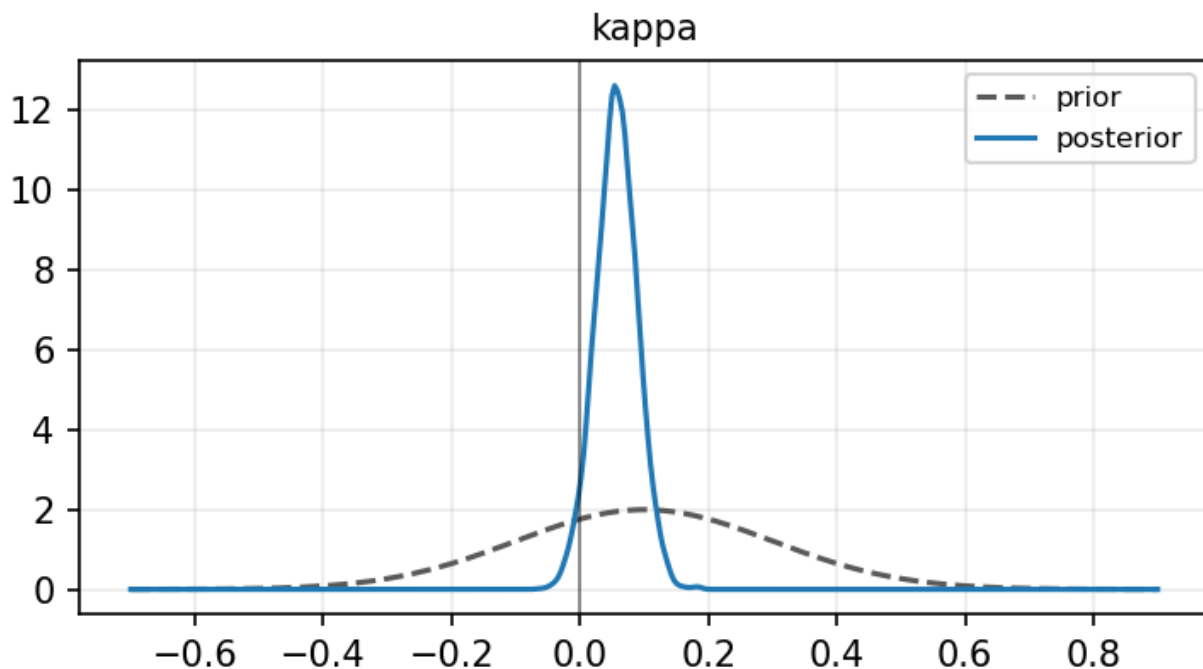
10.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.645 [0.538, 0.749]	0.639 [0.534, 0.712]	0.594 [0.476, 0.711]	0.594 [0.465, 0.715]
kappa	0.057 [-0.008, 0.120]	NaN	0.081 [-0.014, 0.190]	NaN
kappa_0	NaN	-0.020 [-0.427, 0.286]	NaN	0.092 [-0.007, 0.189]
delta	NaN	0.000 [-0.001, 0.003]	NaN	0.011 [-0.034, 0.052]
theta	NaN	NaN	0.220 [-0.285, 0.627]	NaN
theta_0	NaN	NaN	NaN	0.213 [-0.101, 0.531]
gamma	NaN	NaN	NaN	-0.004 [-0.043, 0.035]
rho_1	NaN	1.828 [1.713, 1.920]	0.911 [0.592, 1.138]	0.990 [0.753, 1.218]
rho_2	NaN	-0.832 [-0.921, -0.719]	0.012 [-0.194, 0.223]	-0.048 [-0.266, 0.171]
phi_1	0.950 [0.896, 1.004]	0.950 [0.900, 1.013]	0.949 [0.897, 1.001]	0.950 [0.894, 1.006]
lambda_ez	0.042 [-0.166, 0.247]	0.058 [-0.163, 0.242]	NaN	0.014 [-0.190, 0.220]
n	NaN	-0.092 [-0.194, 0.074]	-0.016 [-0.069, 0.029]	-0.024 [-0.095, 0.039]

10.2 Prior vs Posterior by Model

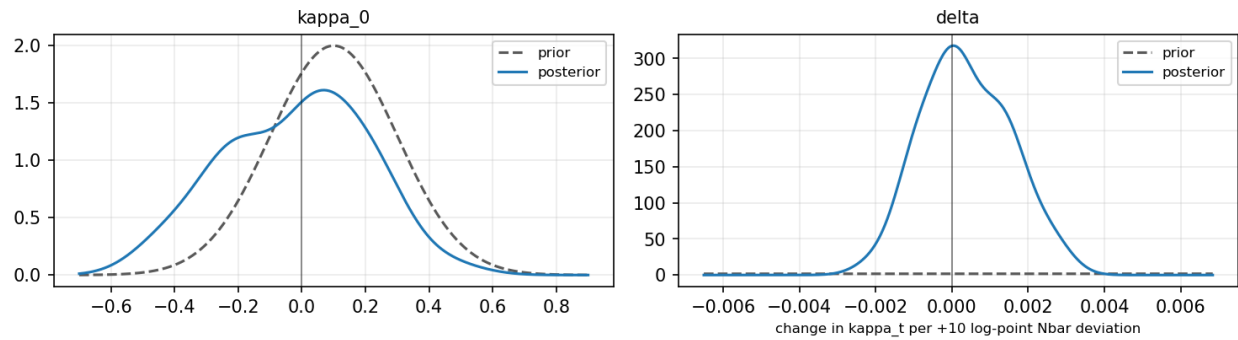
ces

Prior vs posterior — ces



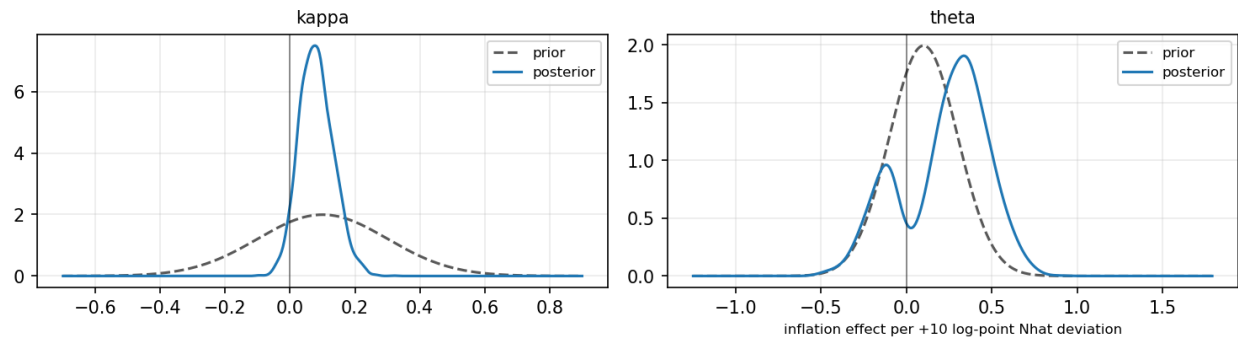
hsa_steady

Prior vs posterior — hsa_steady



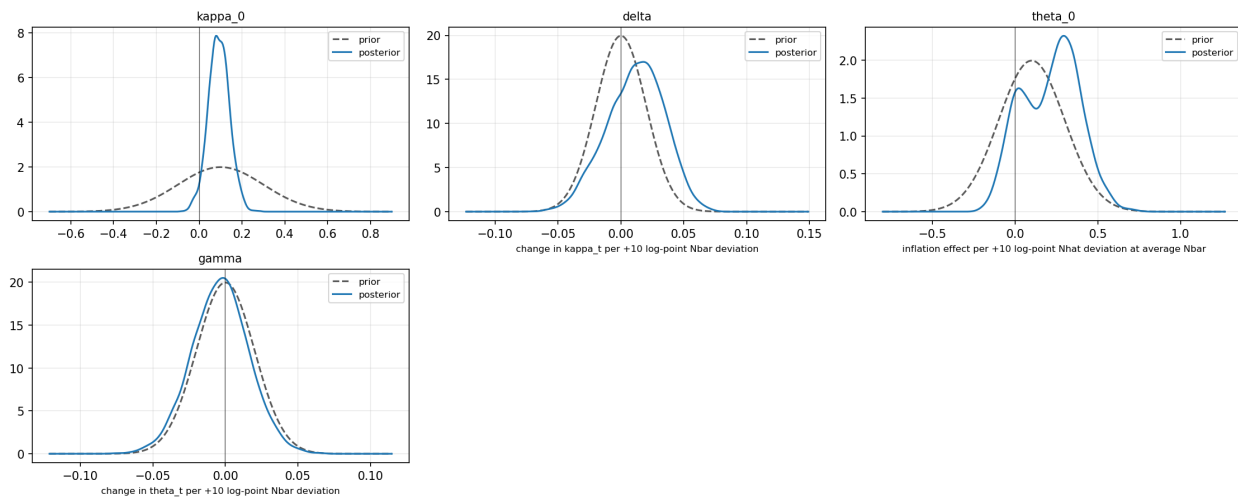
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

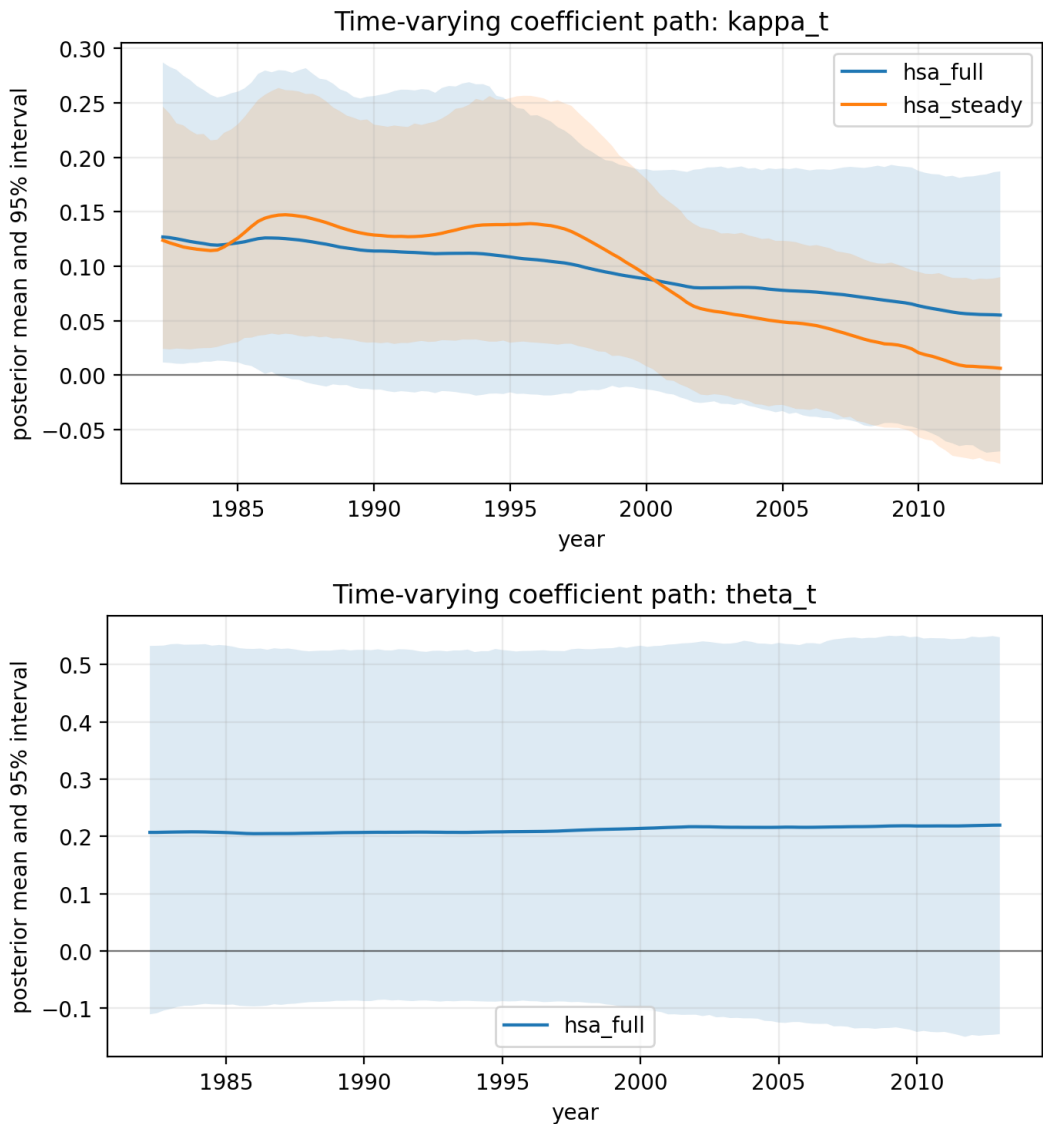
Prior vs posterior — hsa_full



10.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	1.4774
hsa_dynamic	kappa=0	0.1000	0.2000	1.3135
hsa_dynamic	theta=0	0.1000	0.2000	0.2588
hsa_full	kappa_0=0	0.1000	0.2000	0.7075
hsa_full	delta=0	0.0000	0.0200	0.6725
hsa_full	theta_0=0	0.1000	0.2000	0.8982
hsa_full	gamma=0	0.0000	0.0200	1.0219
hsa_steady	kappa_0=0	0.1000	0.2000	0.4520
hsa_steady	delta=0	0.0000	0.0200	0.1758

10.4 Time-Varying Kappa Path



10.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-119.9734	0.6354	-249.0114	1.0000	NaN	NaN	NaN
hsa_dynamic	-115.2706	0.6129	NaN	NaN	NaN	0.2588	NaN
hsa_full	-113.9916	0.6062	NaN	NaN	0.6725	NaN	1.0219
hsa_steady	-115.6751	0.6134	-308.0485	0.0000	0.1758	NaN	NaN

11 unemployment_gap: prior=baseline, period=start_1988

Condition. data=unemployment_gap, prior=baseline, period=start_1988, constraint=unrestricted (unrestricted).

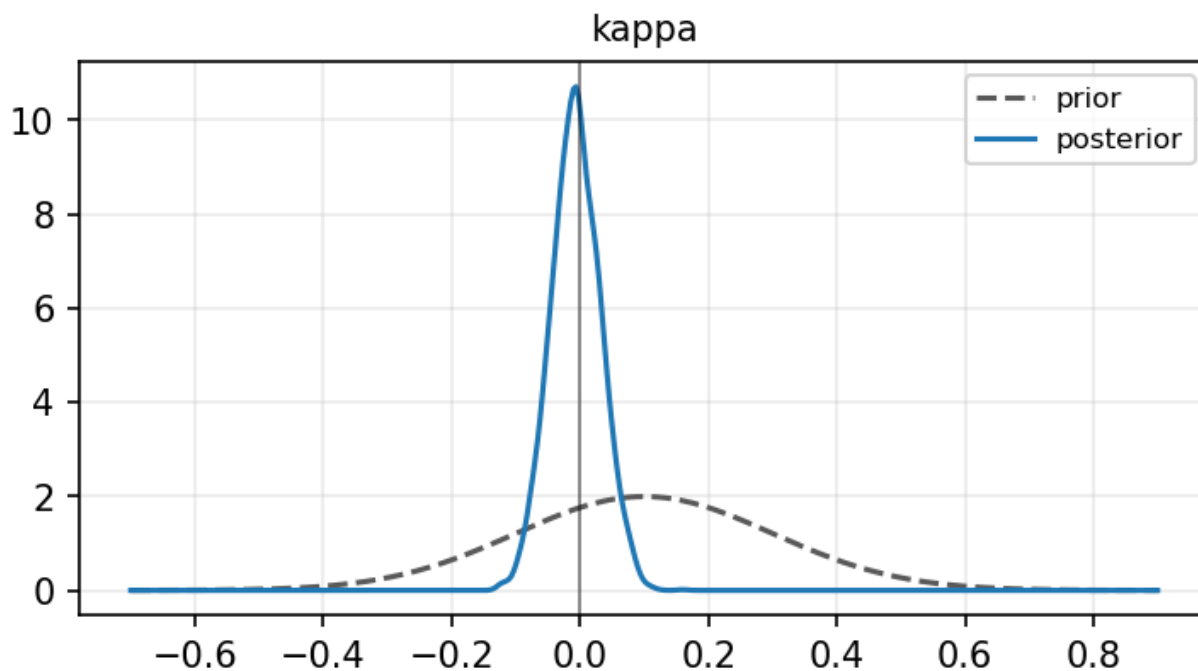
11.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.626 [0.502, 0.746]	0.611 [0.512, 0.695]	0.616 [0.492, 0.736]	0.608 [0.476, 0.731]
kappa	-0.007 [-0.081, 0.070]	NaN	0.000 [-0.074, 0.084]	NaN
kappa_0	NaN	0.053 [-0.469, 0.467]	NaN	0.015 [-0.081, 0.113]
delta	NaN	-0.000 [-0.003, 0.002]	NaN	0.005 [-0.030, 0.037]
theta	NaN	NaN	0.156 [-0.252, 0.555]	NaN
theta_0	NaN	NaN	NaN	0.152 [-0.211, 0.532]
gamma	NaN	NaN	NaN	-0.001 [-0.039, 0.038]
rho_1	NaN	1.697 [1.563, 1.824]	0.668 [0.250, 1.029]	0.856 [0.560, 1.123]
rho_2	NaN	-0.704 [-0.835, -0.571]	-0.071 [-0.389, 0.193]	-0.073 [-0.337, 0.170]
phi_1	0.952 [0.889, 1.014]	0.961 [0.904, 1.015]	0.952 [0.890, 1.013]	0.952 [0.888, 1.016]
lambda_ez	0.068 [-0.166, 0.306]	0.057 [-0.102, 0.290]	NaN	0.063 [-0.174, 0.297]
n	NaN	-0.297 [-0.440, -0.137]	-0.041 [-0.085, 0.002]	-0.041 [-0.102, 0.018]

11.2 Prior vs Posterior by Model

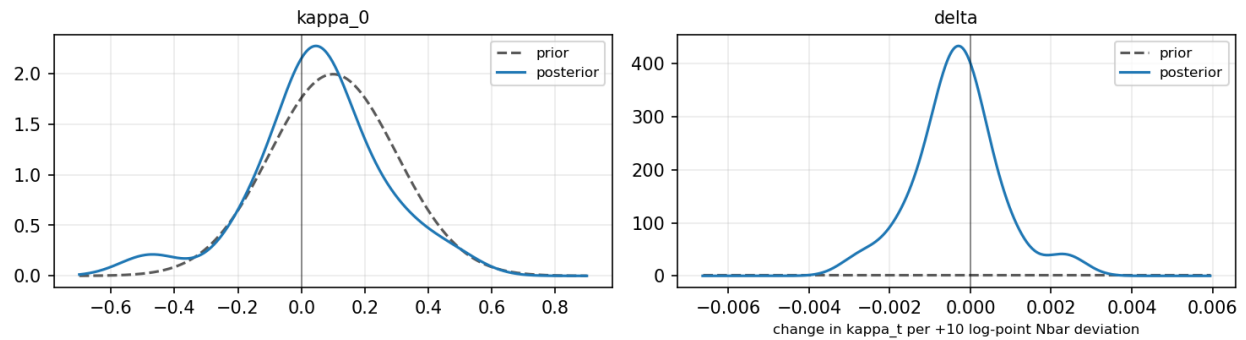
ces

Prior vs posterior — ces



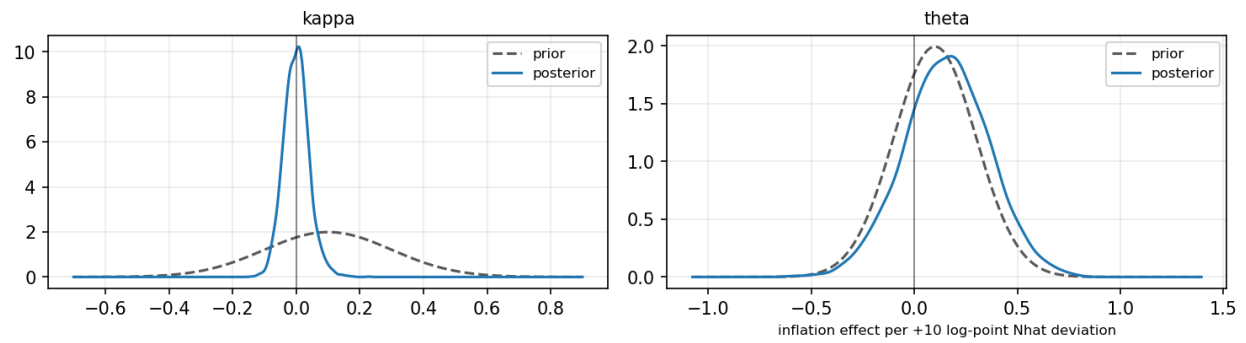
hsa_steady

Prior vs posterior — hsa_steady



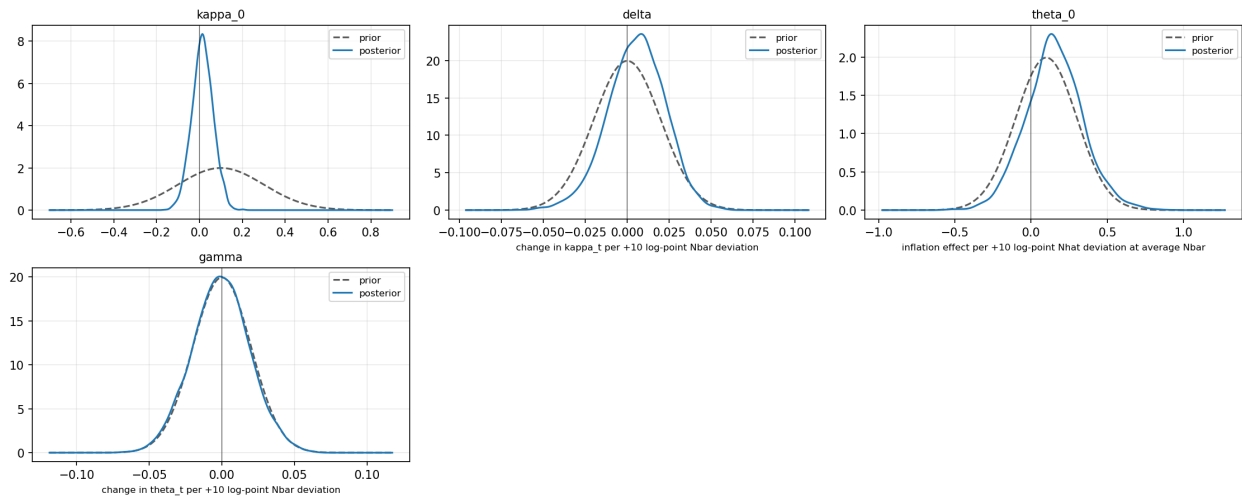
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

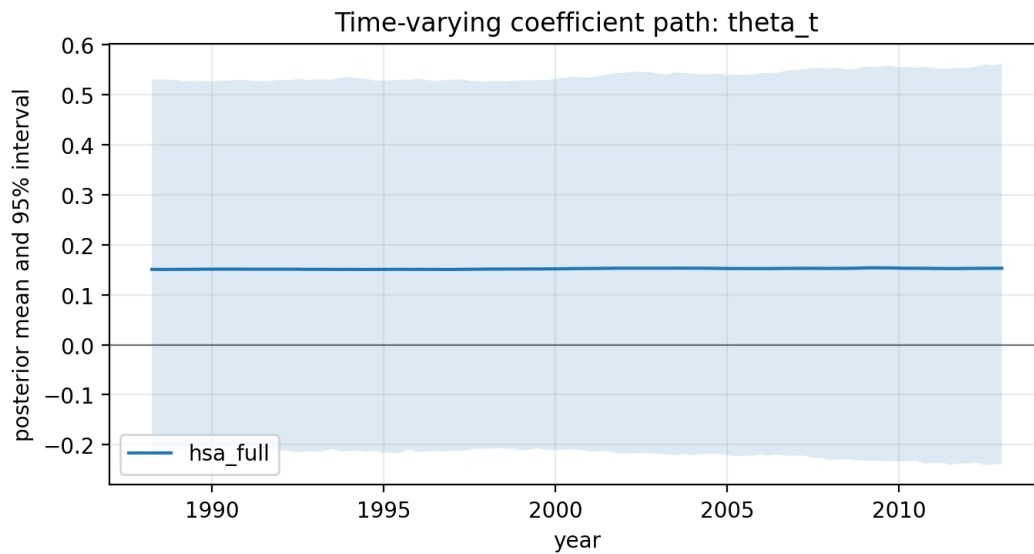
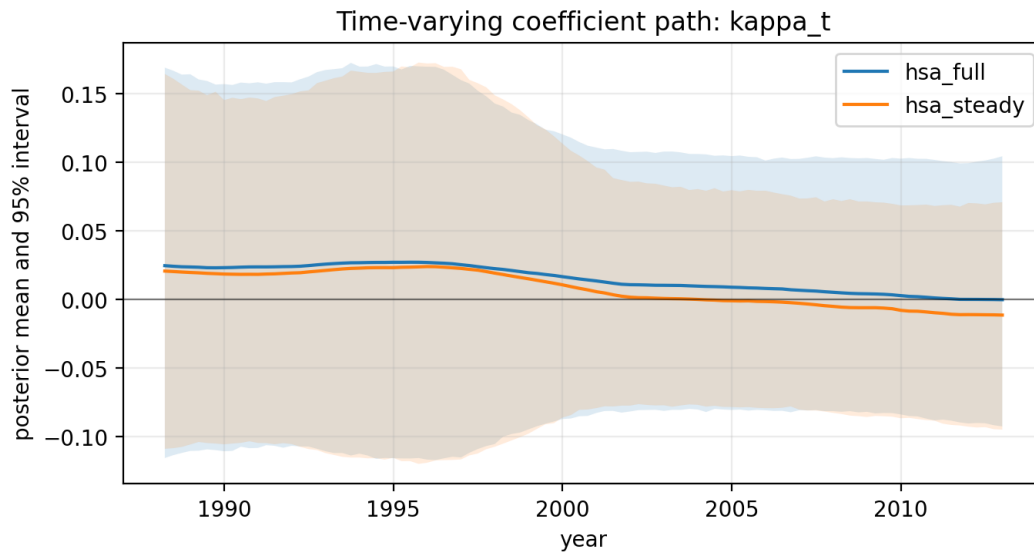
Prior vs posterior — hsa_full



11.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	5.8163
hsa_dynamic	kappa=0	0.1000	0.2000	5.6899
hsa_dynamic	theta=0	0.1000	0.2000	0.8264
hsa_full	kappa_0=0	0.1000	0.2000	4.4689
hsa_full	delta=0	0.0000	0.0200	1.0855
hsa_full	theta_0=0	0.1000	0.2000	0.8152
hsa_full	gamma=0	0.0000	0.0200	1.0027
hsa_steady	kappa_0=0	0.1000	0.2000	4.7279
hsa_steady	delta=0	0.0000	0.0200	1.1336

11.4 Time-Varying Kappa Path



11.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-94.9099	0.6231	-197.6972	1.0000	NaN	NaN	NaN
hsa_dynamic	-93.6367	0.6172	NaN	NaN	NaN	0.8264	NaN
hsa_full	-93.3943	0.6138	NaN	NaN	1.0855	NaN	1.0027
hsa_steady	-94.6200	0.6213	-257.9298	0.0000	1.1336	NaN	NaN

12 unemployment_gap: prior=baseline, period=end_2019

Condition. data=unemployment_gap, prior=baseline, period=end_2019, constraint=unrestricted (unrestricted).

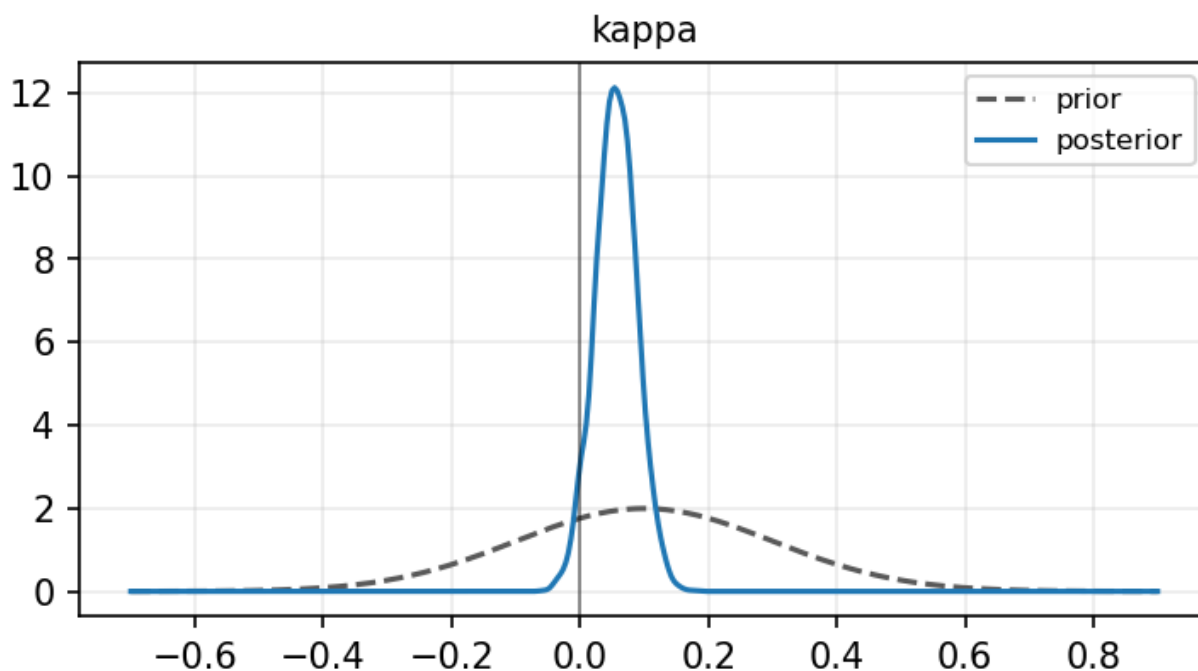
12.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.647 [0.542, 0.749]	0.653 [0.553, 0.757]	0.601 [0.484, 0.713]	0.586 [0.468, 0.698]
kappa	0.056 [-0.007, 0.119]	NaN	0.076 [-0.022, 0.187]	NaN
kappa_0	NaN	0.004 [-0.491, 0.374]	NaN	0.097 [0.008, 0.191]
delta	NaN	0.000 [-0.002, 0.003]	NaN	0.008 [-0.032, 0.048]
theta	NaN	NaN	0.174 [-0.303, 0.603]	NaN
theta_0	NaN	NaN	NaN	0.279 [-0.068, 0.570]
gamma	NaN	NaN	NaN	-0.003 [-0.042, 0.038]
rho_1	NaN	1.823 [1.761, 1.886]	0.917 [0.602, 1.143]	0.971 [0.736, 1.206]
rho_2	NaN	-0.828 [-0.892, -0.770]	0.008 [-0.200, 0.221]	-0.047 [-0.279, 0.170]
phi_1	0.950 [0.896, 1.002]	0.941 [0.889, 0.998]	0.949 [0.897, 1.003]	0.950 [0.894, 1.005]
lambda_ez	0.042 [-0.168, 0.259]	0.030 [-0.130, 0.180]	NaN	0.016 [-0.190, 0.226]
n	NaN	-0.057 [-0.153, 0.025]	-0.019 [-0.074, 0.027]	-0.006 [-0.060, 0.050]

12.2 Prior vs Posterior by Model

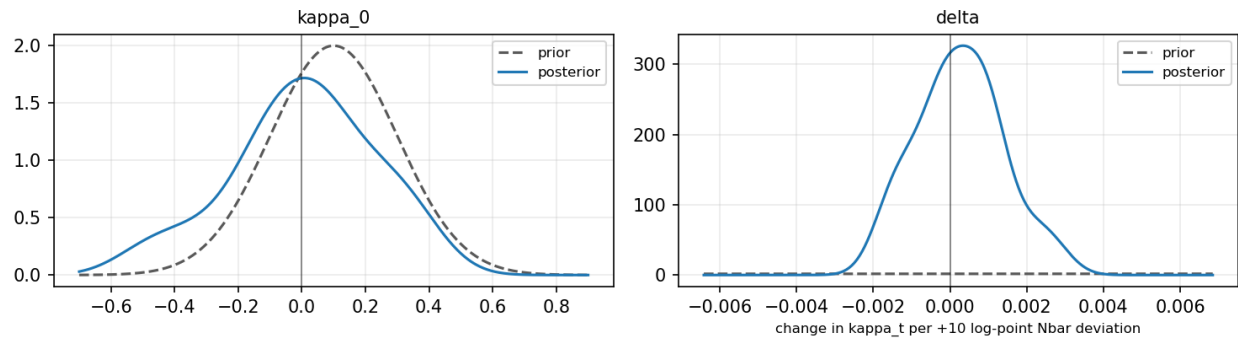
ces

Prior vs posterior — ces



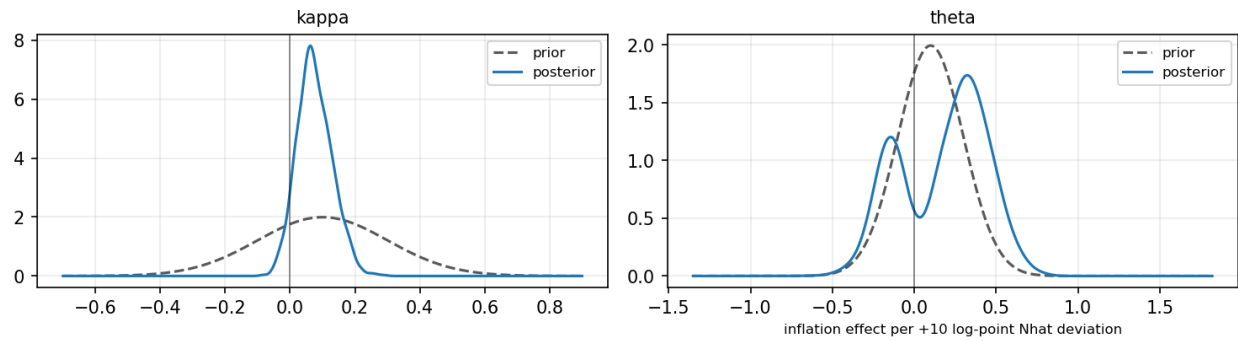
hsa_steady

Prior vs posterior — hsa_steady



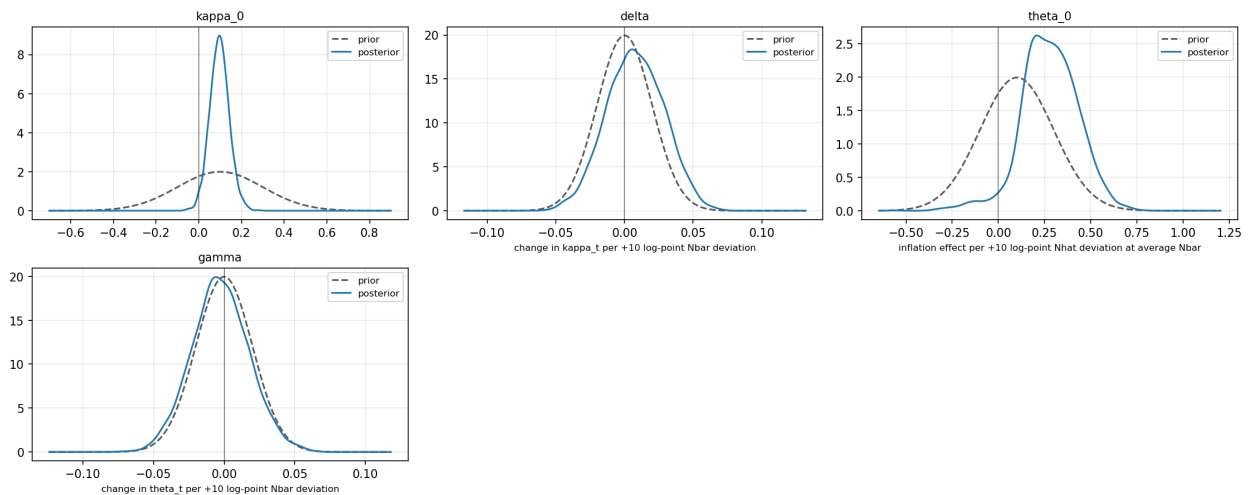
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

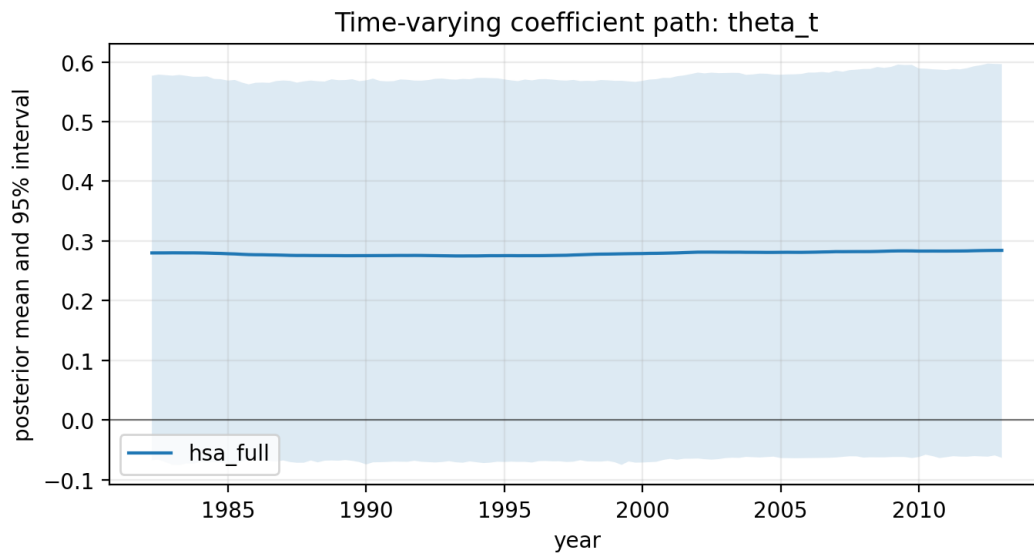
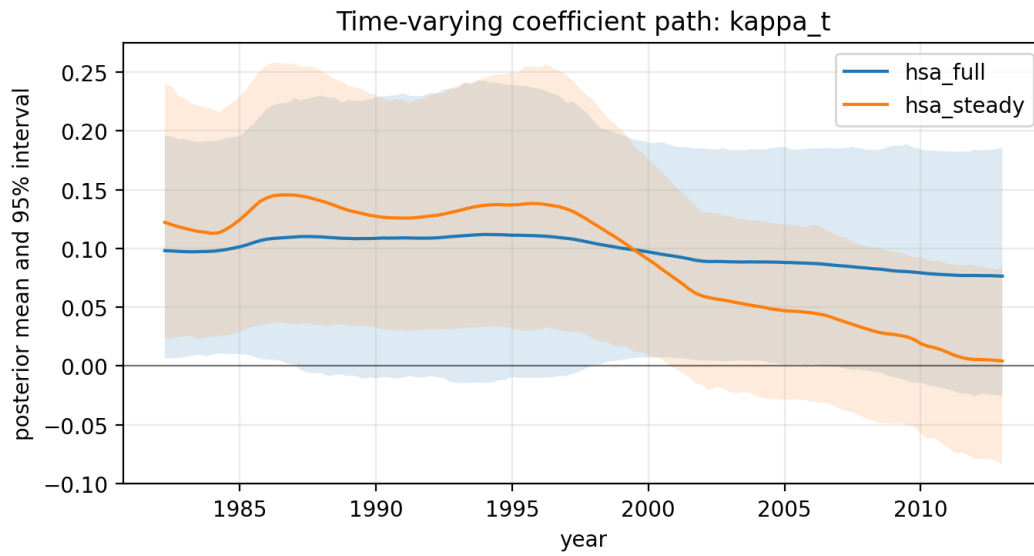
Prior vs posterior — hsa_full



12.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	1.7326
hsa_dynamic	kappa=0	0.1000	0.2000	1.6525
hsa_dynamic	theta=0	0.1000	0.2000	0.3211
hsa_full	kappa_0=0	0.1000	0.2000	0.5671
hsa_full	delta=0	0.0000	0.0200	0.8656
hsa_full	theta_0=0	0.1000	0.2000	0.1556
hsa_full	gamma=0	0.0000	0.0200	0.9667
hsa_steady	kappa_0=0	0.1000	0.2000	0.4913
hsa_steady	delta=0	0.0000	0.0200	0.1503

12.4 Time-Varying Kappa Path



12.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-119.9817	0.6354	-249.0072	1.0000	NaN	NaN	NaN
hsa_dynamic	-117.3150	0.6228	NaN	NaN	NaN	0.3211	NaN
hsa_full	-108.8318	0.5790	NaN	NaN	0.8656	NaN	0.9667
hsa_steady	-115.6317	0.6133	-307.9593	0.0000	0.1503	NaN	NaN

13 unemployment_gap: prior=weak, period=full

Condition. data=unemployment_gap, prior=weak, period=full, constraint=unrestricted (unrestricted).

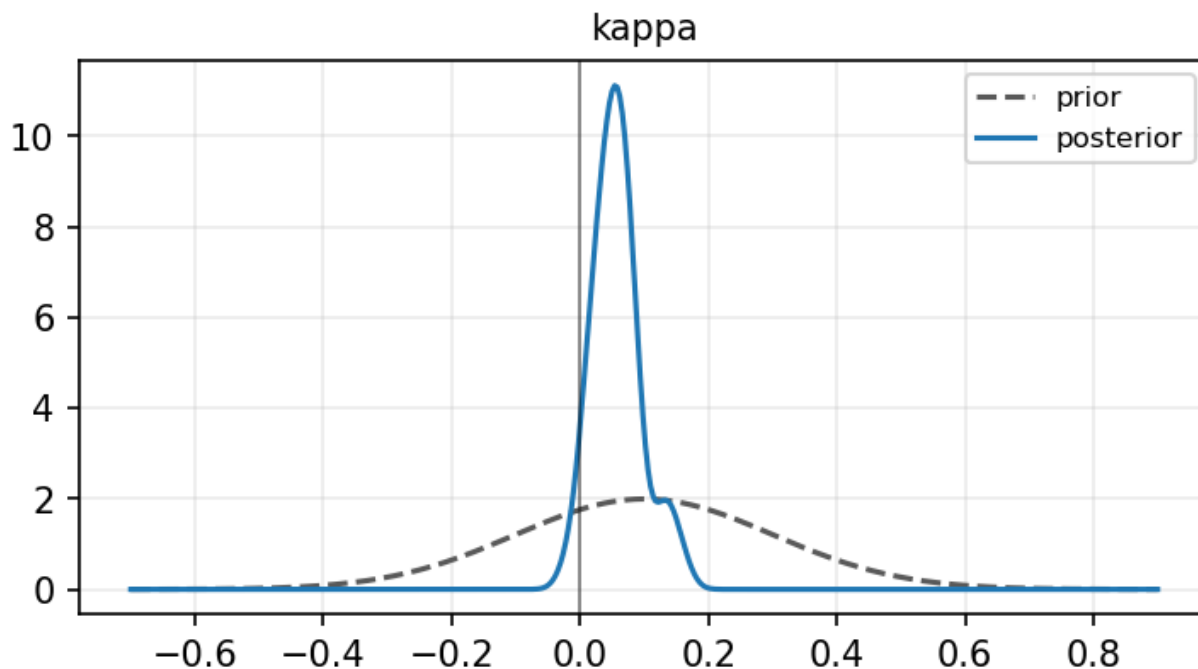
13.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.643 [0.549, 0.762]	0.658 [0.549, 0.758]	0.592 [0.521, 0.663]	0.634 [0.533, 0.744]
kappa	0.058 [0.006, 0.140]	NaN	0.094 [0.044, 0.160]	NaN
kappa_0	NaN	0.076 [-0.367, 0.366]	NaN	-0.059 [-0.370, 0.217]
delta	NaN	-0.000 [-0.002, 0.002]	NaN	0.001 [-0.001, 0.003]
theta	NaN	NaN	0.012 [0.006, 0.018]	NaN
theta_0	NaN	NaN	NaN	0.026 [-0.392, 0.434]
gamma	NaN	NaN	NaN	0.000 [-0.002, 0.002]
rho_1	NaN	1.813 [1.714, 1.901]	1.872 [1.790, 1.941]	1.194 [1.005, 1.392]
rho_2	NaN	-0.817 [-0.905, -0.716]	-0.876 [-0.945, -0.792]	-0.272 [-0.472, -0.068]
phi_1	0.953 [0.891, 0.996]	0.953 [0.907, 0.990]	0.955 [0.912, 1.000]	0.950 [0.915, 1.002]
lambda_ez	0.028 [-0.137, 0.182]	0.023 [-0.148, 0.188]	NaN	0.019 [-0.170, 0.220]
n	NaN	-0.061 [-0.175, 0.065]	0.001 [-0.091, 0.097]	-0.090 [-0.181, 0.037]

13.2 Prior vs Posterior by Model

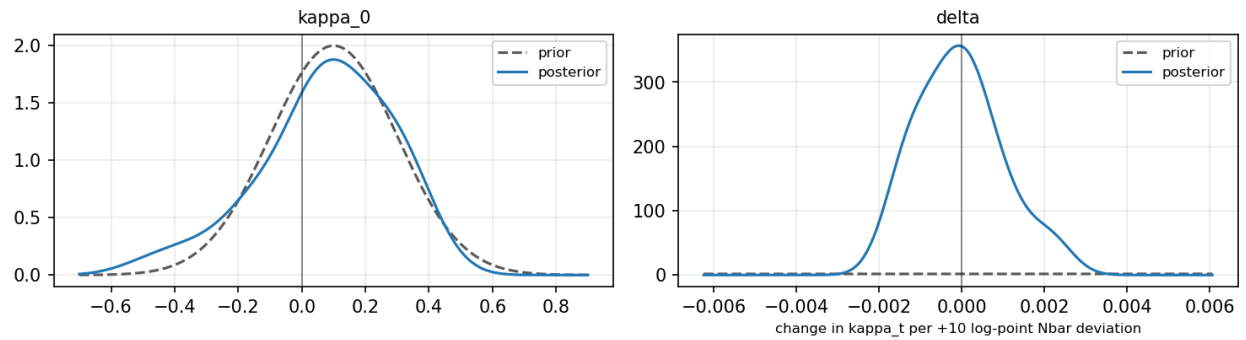
ces

Prior vs posterior — ces



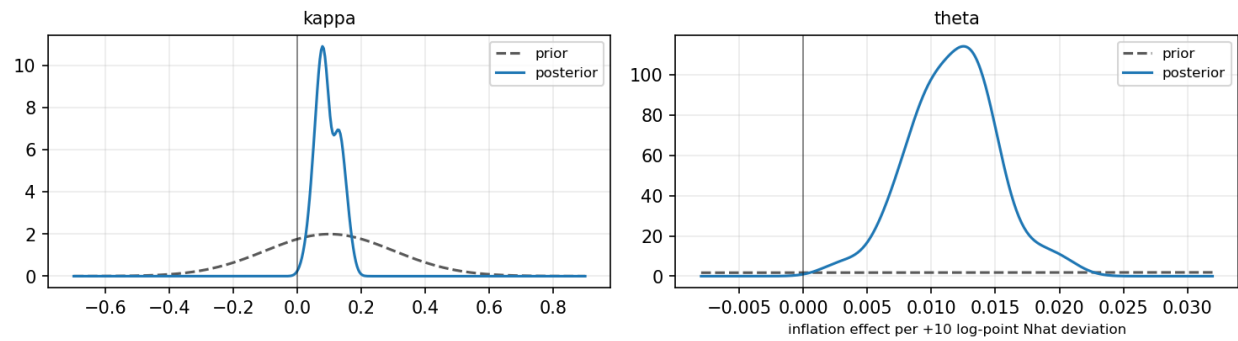
hsa_steady

Prior vs posterior — hsa_steady



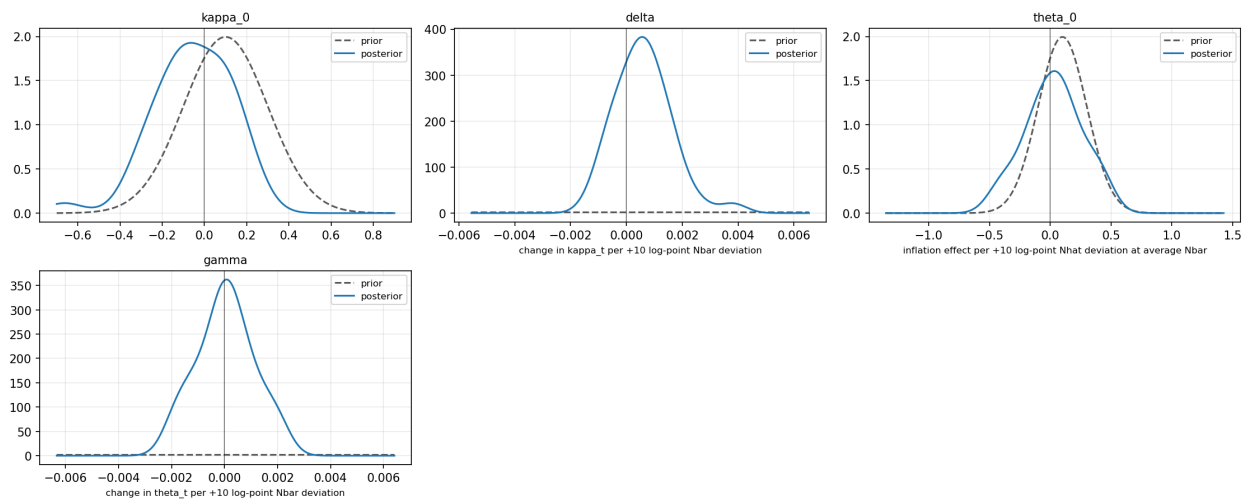
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

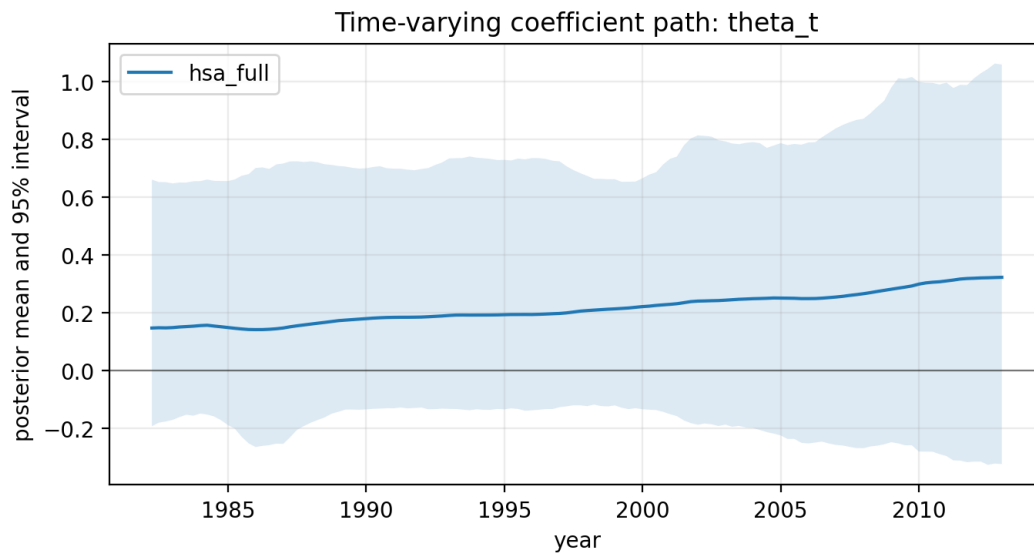
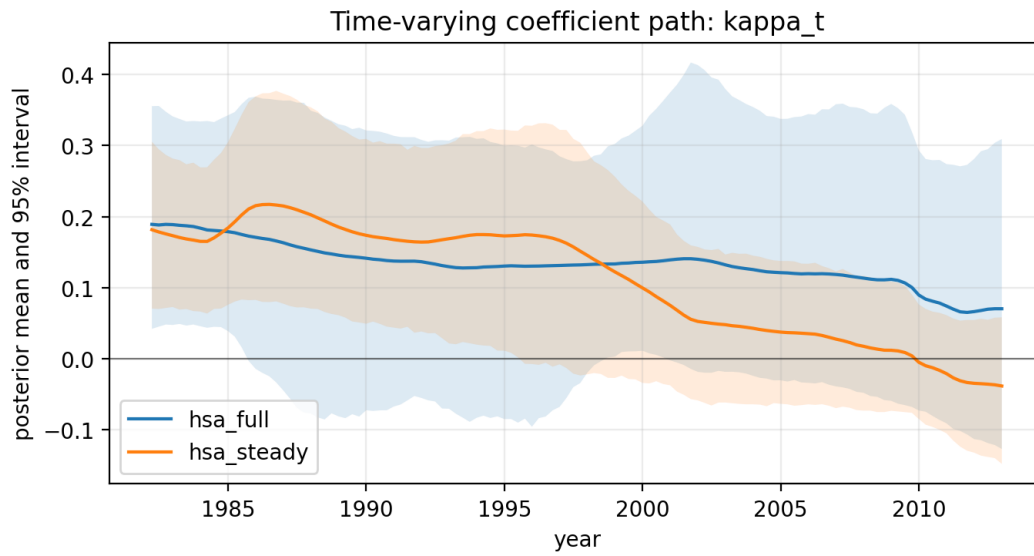
Prior vs posterior — hsa_full



13.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.5000	3.6152
hsa_dynamic	kappa=0	0.1000	0.5000	2.3786
hsa_dynamic	theta=0	0.0000	0.5000	0.2696
hsa_full	kappa_0=0	0.1000	0.5000	1.1080
hsa_full	delta=0	0.0000	0.1000	0.9789
hsa_full	theta_0=0	0.0000	0.5000	1.1208
hsa_full	gamma=0	0.0000	0.1000	0.8705
hsa_steady	kappa_0=0	0.1000	0.5000	1.3149
hsa_steady	delta=0	0.0000	0.1000	0.1221

13.4 Time-Varying Kappa Path



13.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-119.8622	0.6354	-249.0496	1.0000	NaN	NaN	NaN
hsa_dynamic	-110.3062	0.5888	NaN	NaN	NaN	0.2696	NaN
hsa_full	-109.6900	0.5851	NaN	NaN	0.9789	NaN	0.8705
hsa_steady	-113.4699	0.6031	-299.8493	0.0000	0.1221	NaN	NaN

14 unemployment_gap: prior=tight, period=full

Condition. data=unemployment_gap, prior=tight, period=full, constraint=unrestricted (unrestricted).

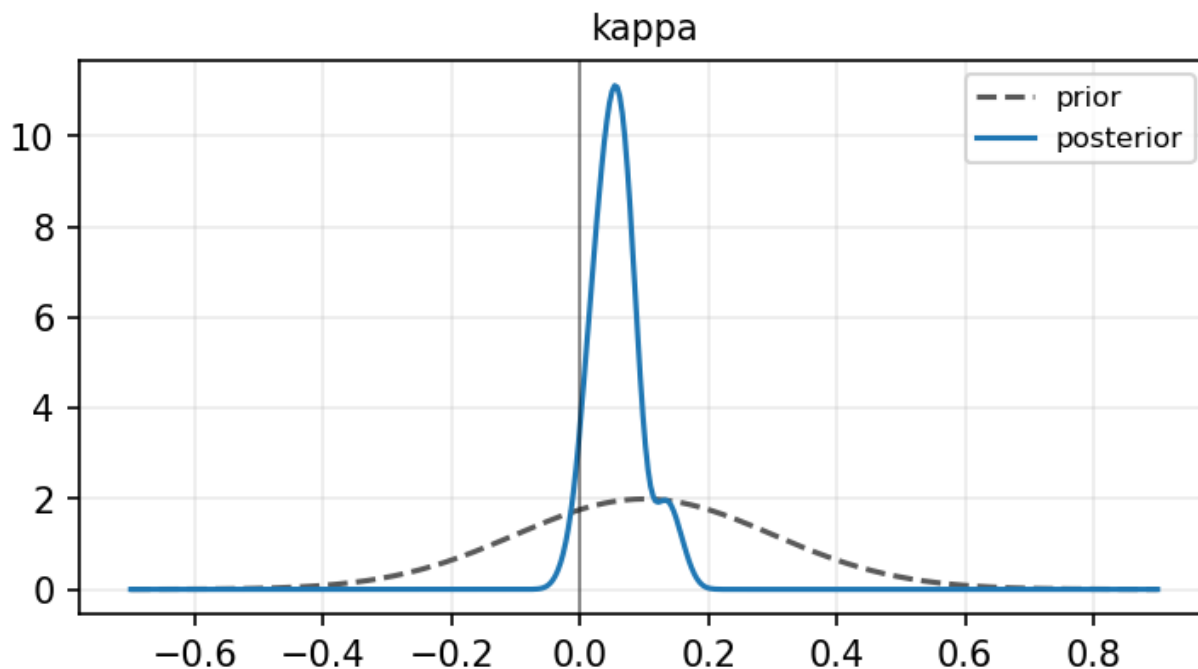
14.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.643 [0.549, 0.762]	0.658 [0.549, 0.758]	0.592 [0.521, 0.663]	0.634 [0.533, 0.744]
kappa	0.058 [0.006, 0.140]	NaN	0.094 [0.044, 0.160]	NaN
kappa_0	NaN	0.076 [-0.367, 0.366]	NaN	-0.059 [-0.370, 0.217]
delta	NaN	-0.000 [-0.002, 0.002]	NaN	0.001 [-0.001, 0.003]
theta	NaN	NaN	0.012 [0.006, 0.018]	NaN
theta_0	NaN	NaN	NaN	0.026 [-0.392, 0.434]
gamma	NaN	NaN	NaN	0.000 [-0.002, 0.002]
rho_1	NaN	1.813 [1.714, 1.901]	1.872 [1.790, 1.941]	1.194 [1.005, 1.392]
rho_2	NaN	-0.817 [-0.905, -0.716]	-0.876 [-0.945, -0.792]	-0.272 [-0.472, -0.068]
phi_1	0.953 [0.891, 0.996]	0.953 [0.907, 0.990]	0.955 [0.912, 1.000]	0.950 [0.915, 1.002]
lambda_ez	0.028 [-0.137, 0.182]	0.023 [-0.148, 0.188]	NaN	0.019 [-0.170, 0.220]
n	NaN	-0.061 [-0.175, 0.065]	0.001 [-0.091, 0.097]	-0.090 [-0.181, 0.037]

14.2 Prior vs Posterior by Model

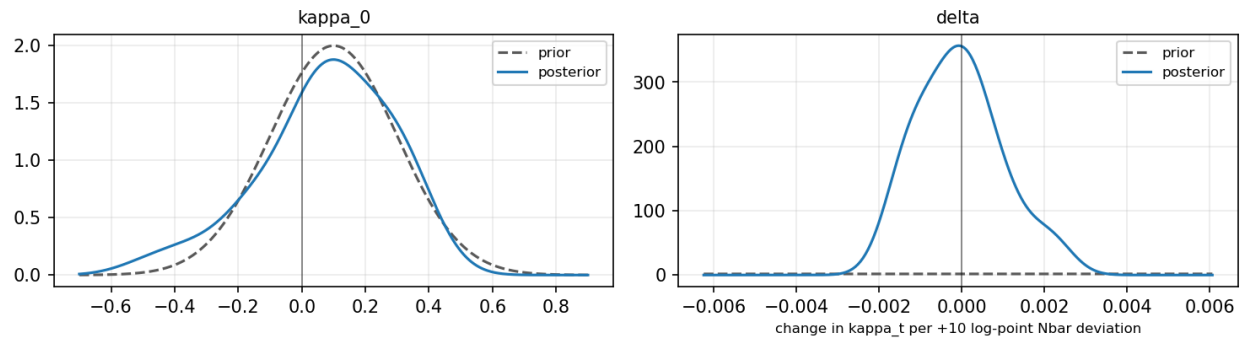
ces

Prior vs posterior — ces



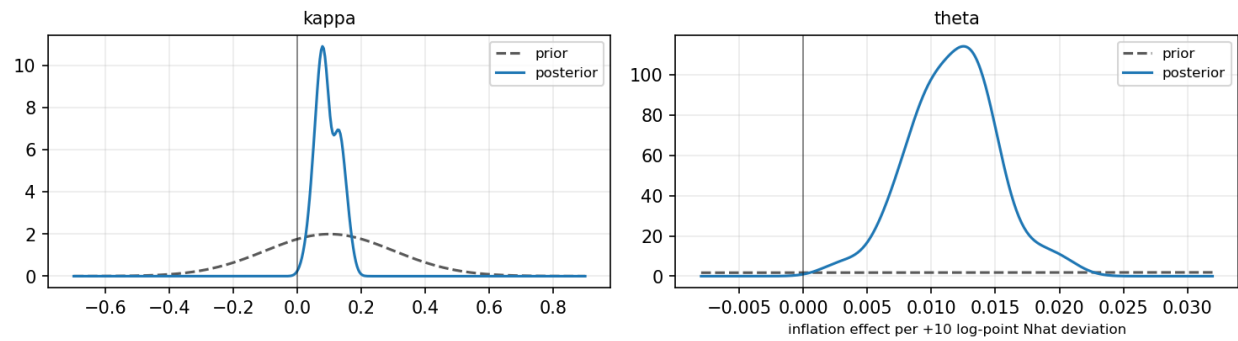
hsa_steady

Prior vs posterior — hsa_steady



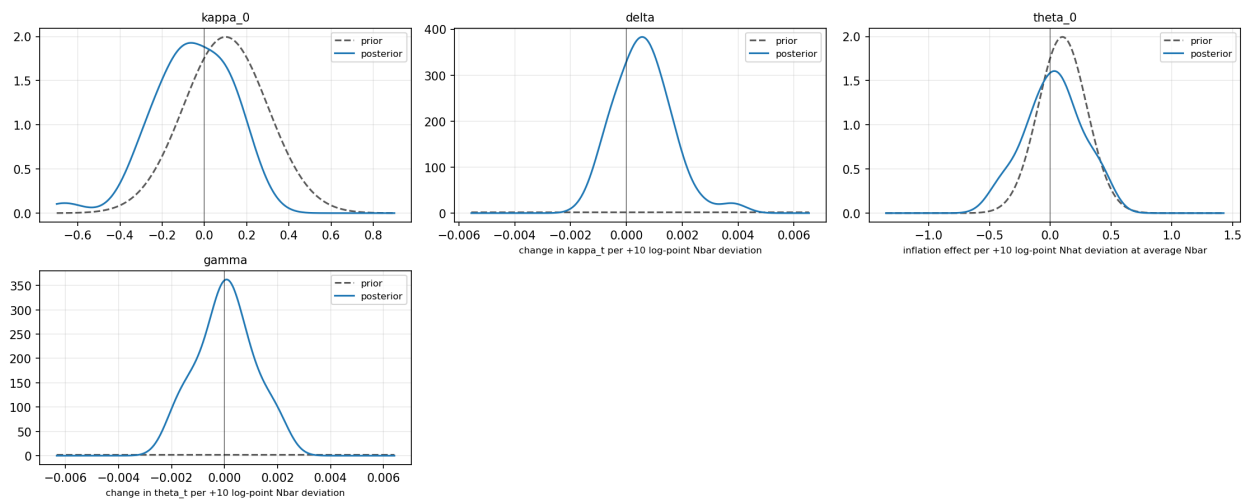
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

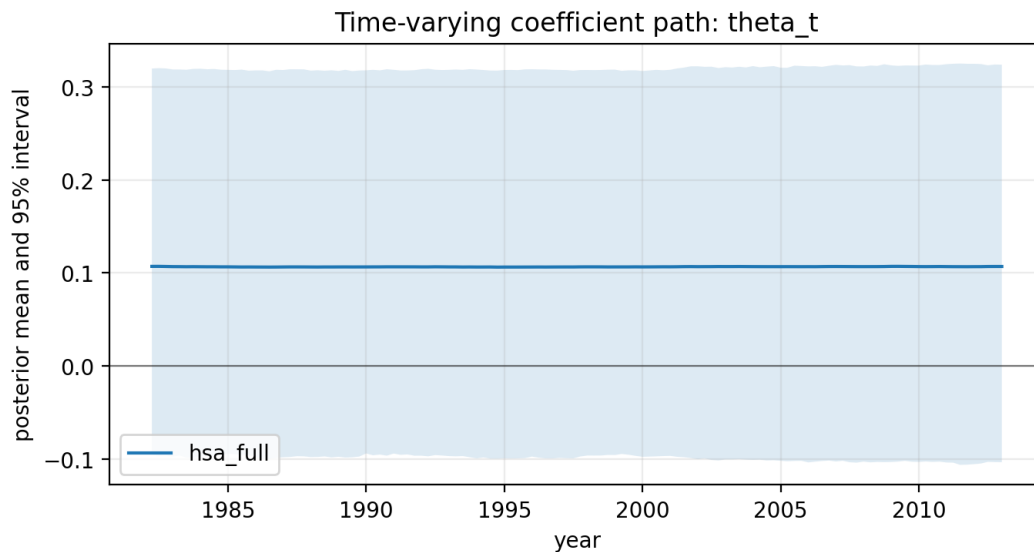
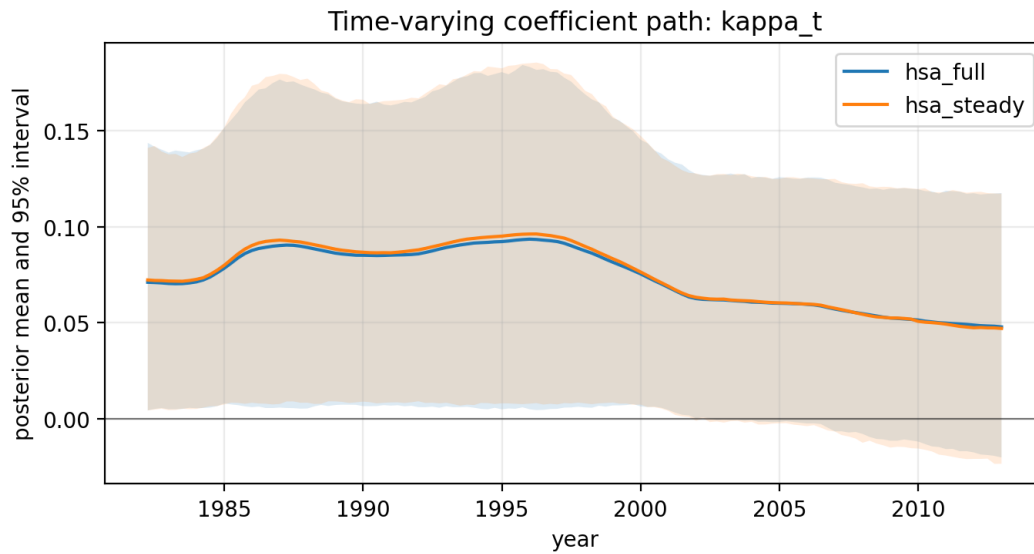
Prior vs posterior — hsa_full



14.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.1000	0.9105
hsa_dynamic	kappa=0	0.1000	0.1000	0.6258
hsa_dynamic	theta=0	0.1000	0.1000	0.9504
hsa_full	kappa_0=0	0.1000	0.1000	0.5642
hsa_full	delta=0	0.0000	0.0100	0.6317
hsa_full	theta_0=0	0.1000	0.1000	0.9528
hsa_full	gamma=0	0.0000	0.0100	1.0321
hsa_steady	kappa_0=0	0.1000	0.1000	0.4973
hsa_steady	delta=0	0.0000	0.0100	0.6218

14.4 Time-Varying Kappa Path



14.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-120.4083	0.6364	-248.9651	1.0000	NaN	NaN	NaN
hsa_dynamic	-119.8150	0.6357	NaN	NaN	NaN	0.9504	NaN
hsa_full	-118.8726	0.6285	NaN	NaN	0.6317	NaN	1.0321
hsa_steady	-119.1432	0.6298	-335.6280	0.0000	0.6218	NaN	NaN

15 Notes

WAIC and LOO are not used as primary criteria because the models contain latent states. Chib marginal likelihoods are reported only when the legacy ordinate calculation can be matched to the sampler units and supplied data.